

BFS Traversal

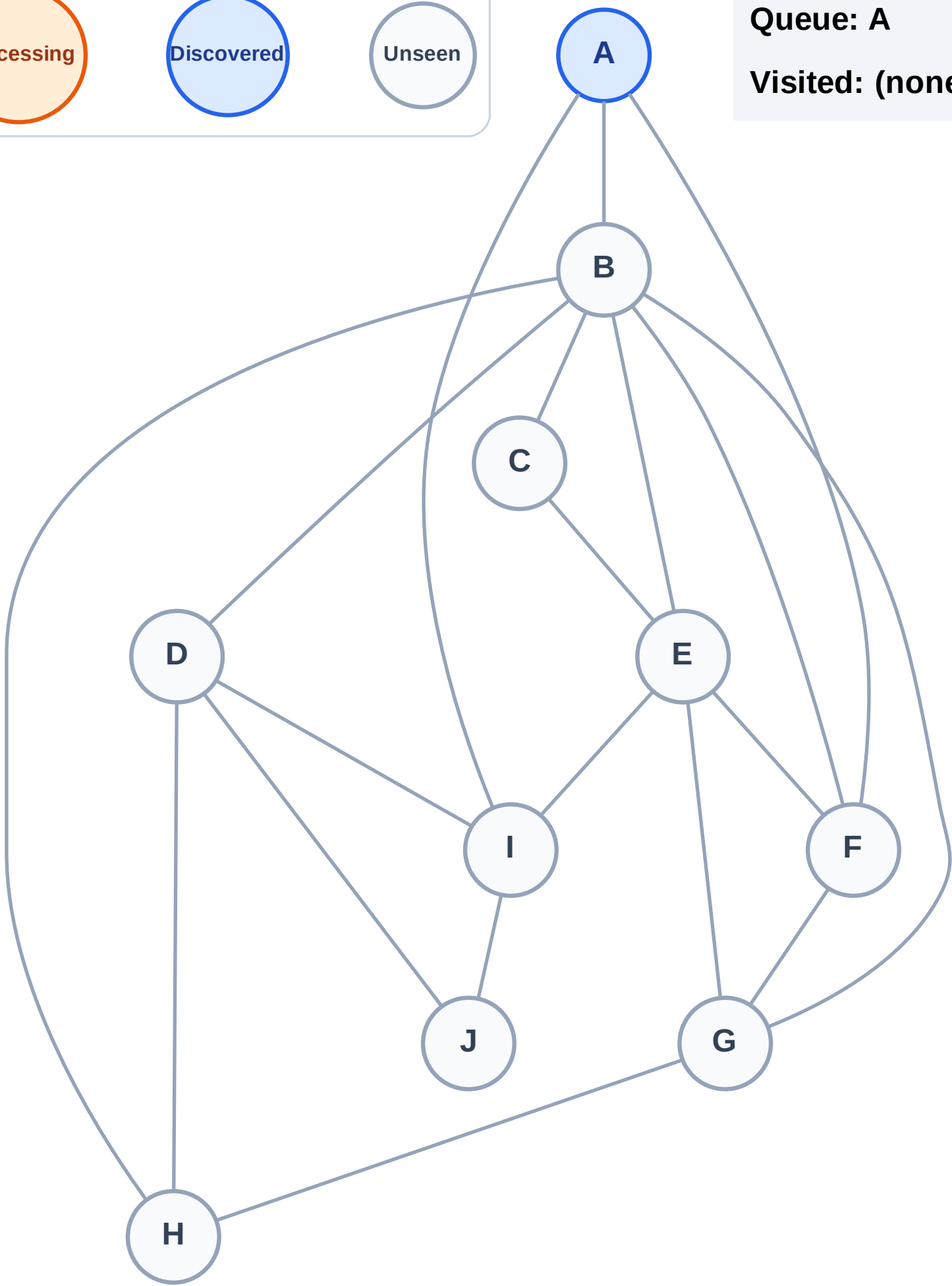
Step 1: Start at A

Legend



Queue: A

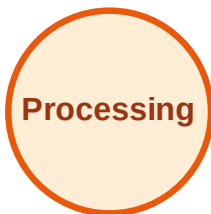
Visited: (none)



BFS Traversal

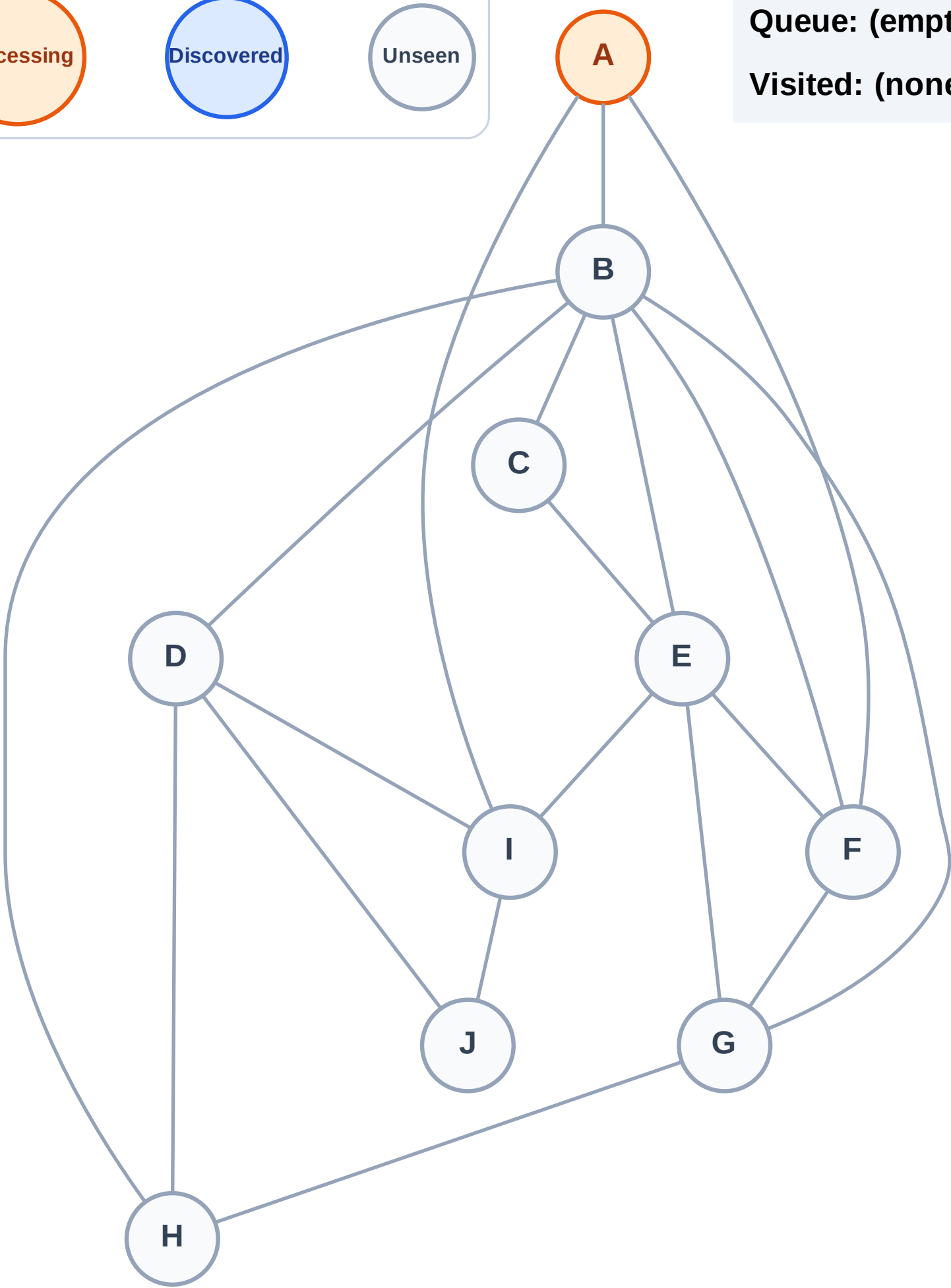
Step 2: Process node A

Legend



Queue: (empty)

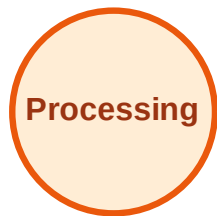
Visited: (none)



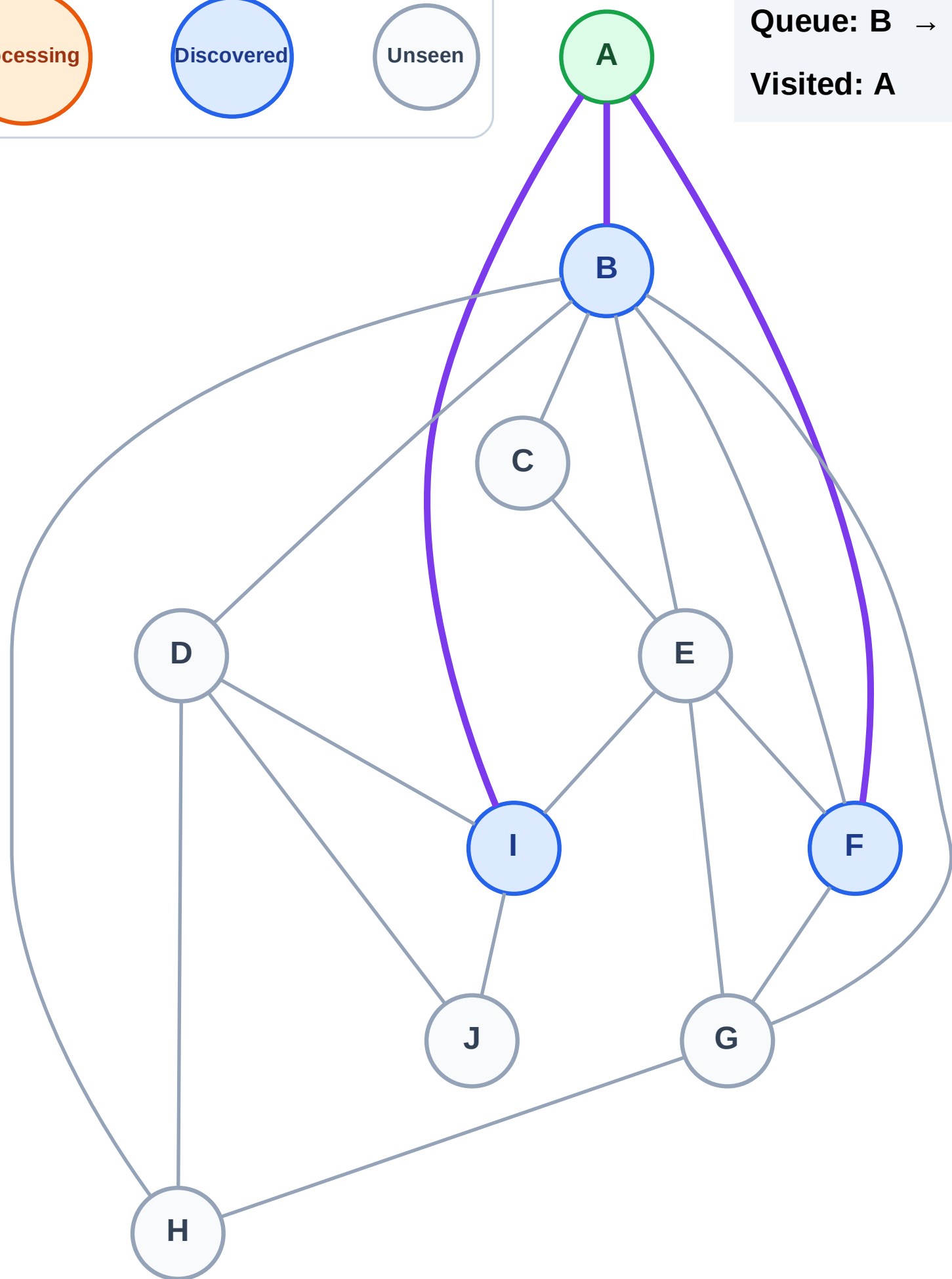
BFS Traversal

Step 3: Discover B, F, I from A

Legend



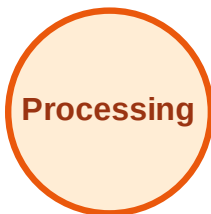
Queue: B → F → I
Visited: A



BFS Traversal

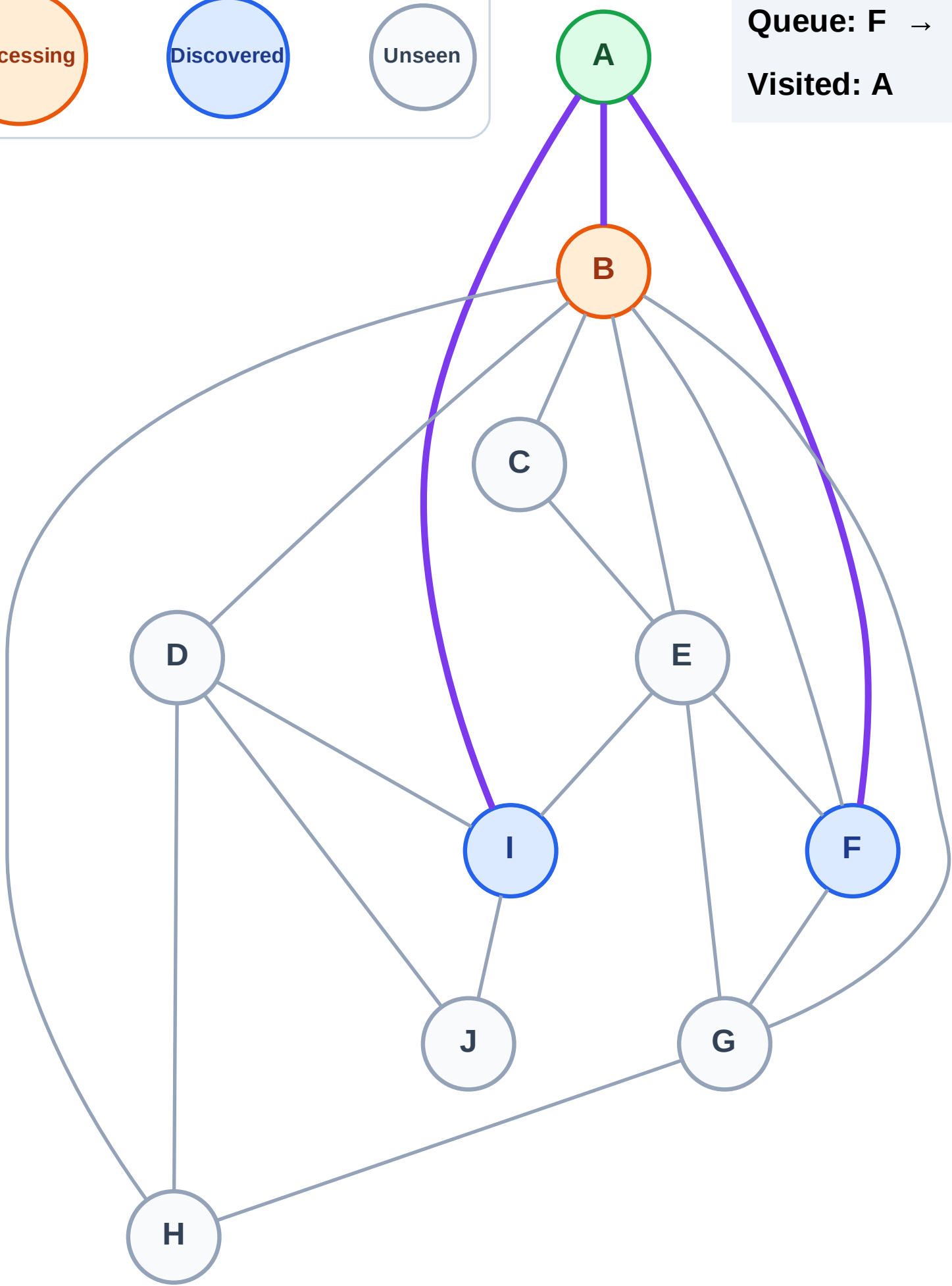
Step 4: Process node B

Legend



Queue: F → I

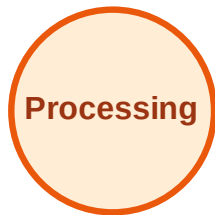
Visited: A



BFS Traversal

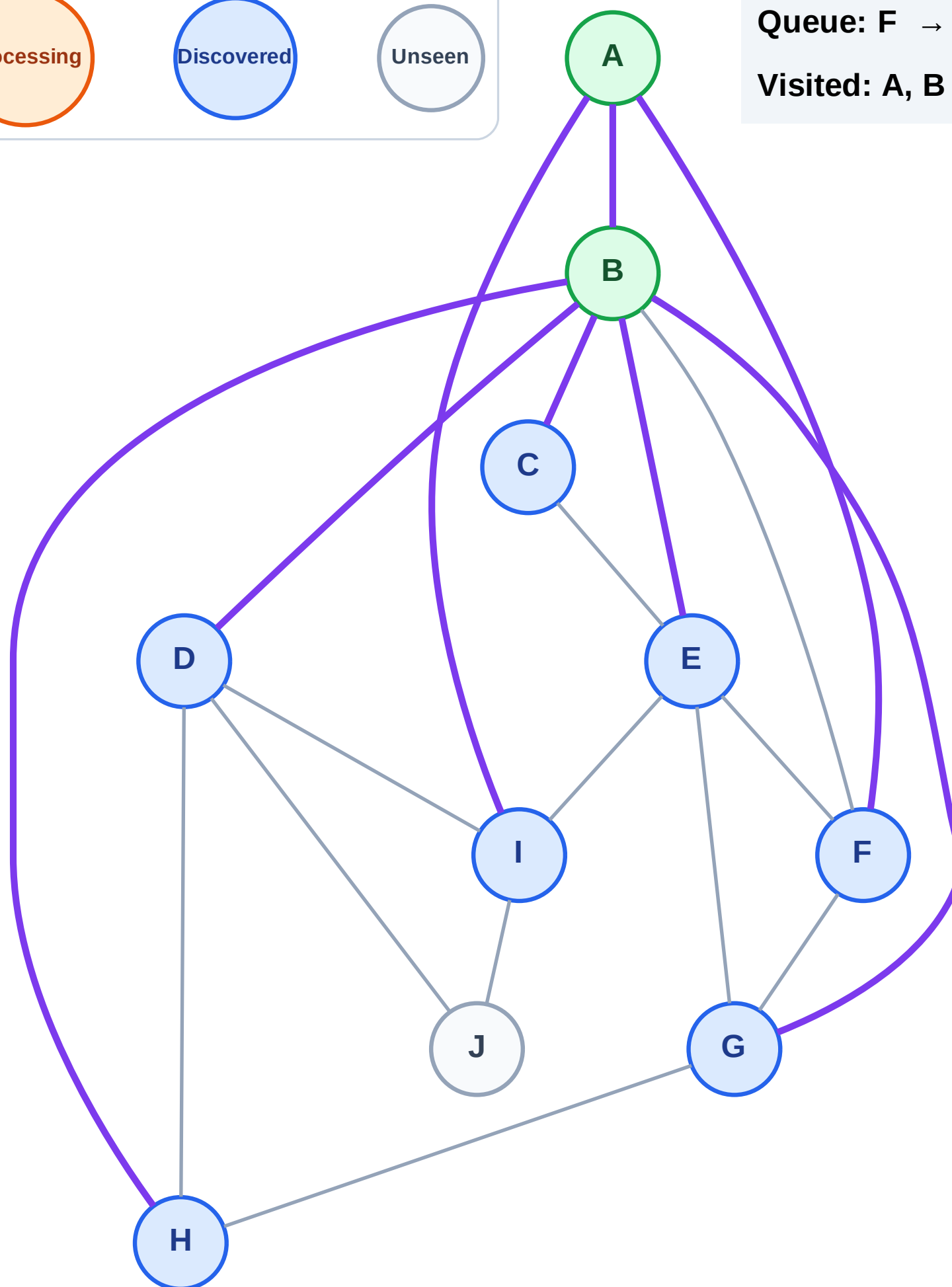
Step 5: Discover C, D, E, G, H from B

Legend



Queue: F → I → C → D → E → G → H

Visited: A, B



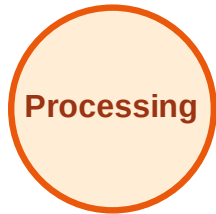
BFS Traversal

Step 6: Process node F

Legend



Complete



Processing



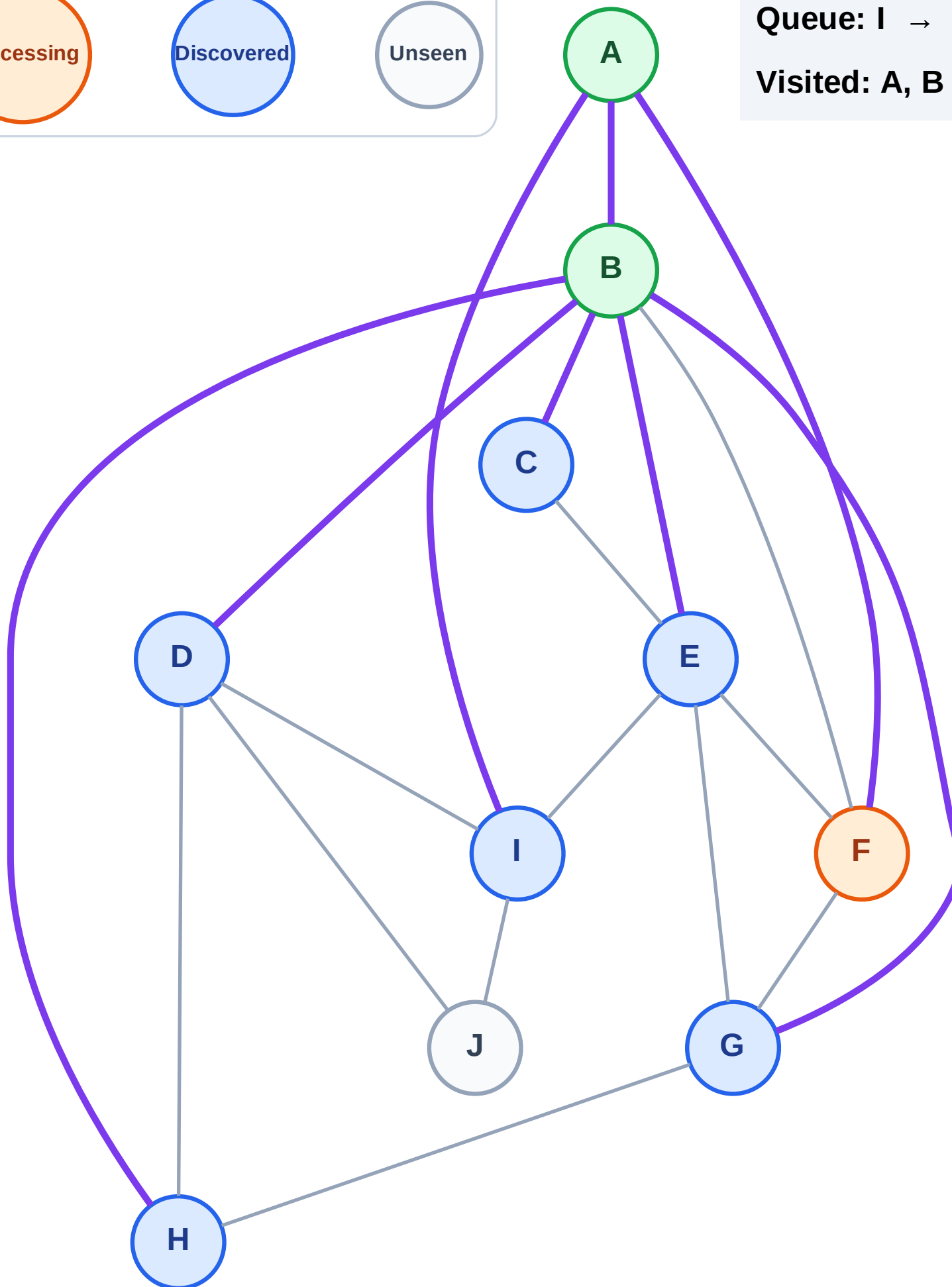
Discovered



Unseen

Queue: I → C → D → E → G → H

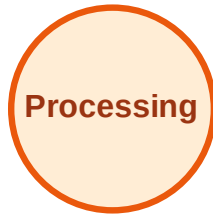
Visited: A, B



BFS Traversal

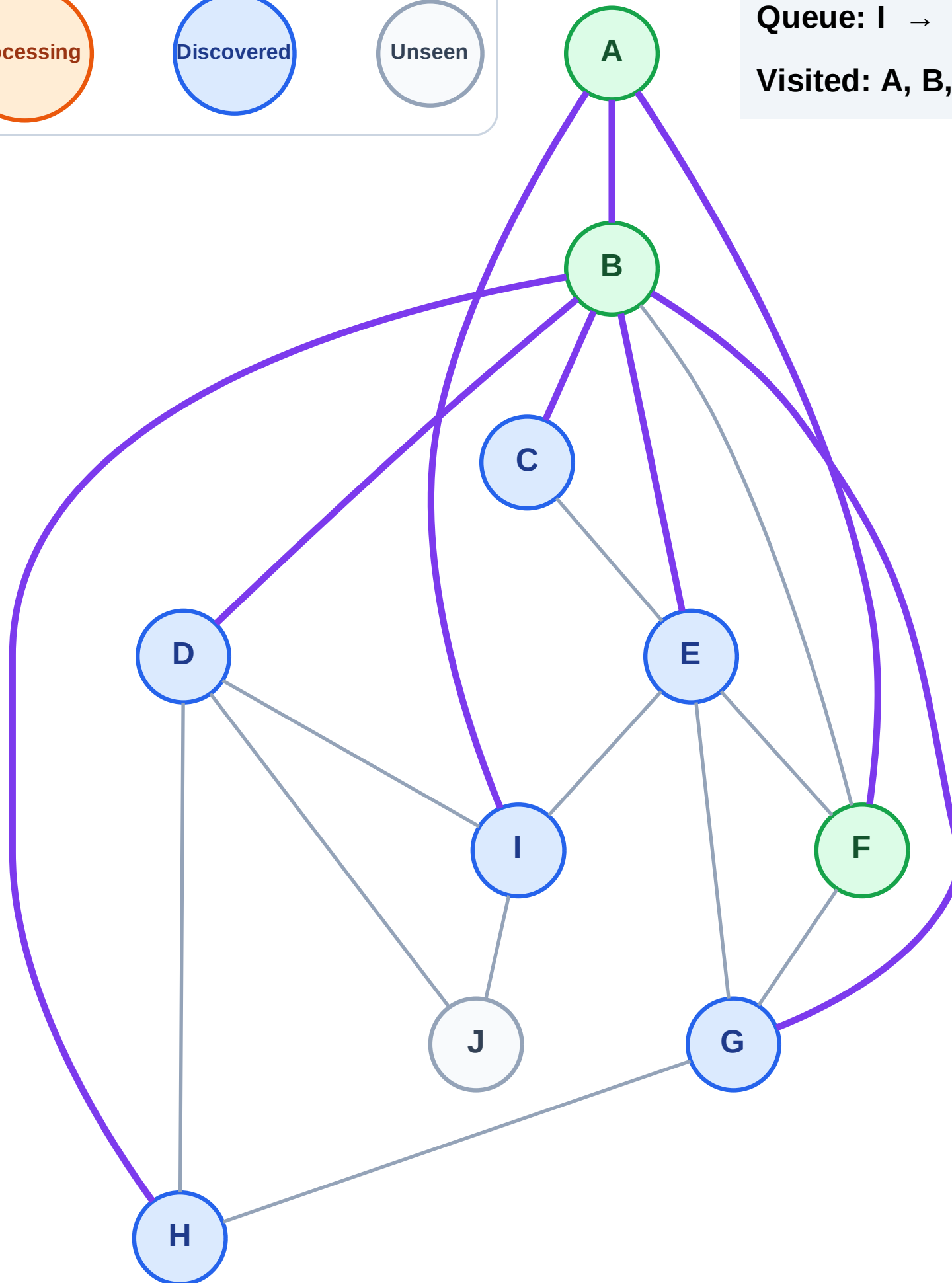
Step 7: No new neighbors from F

Legend



Queue: I → C → D → E → G → H

Visited: A, B, F



BFS Traversal

Step 8: Process node I

Legend

Complete

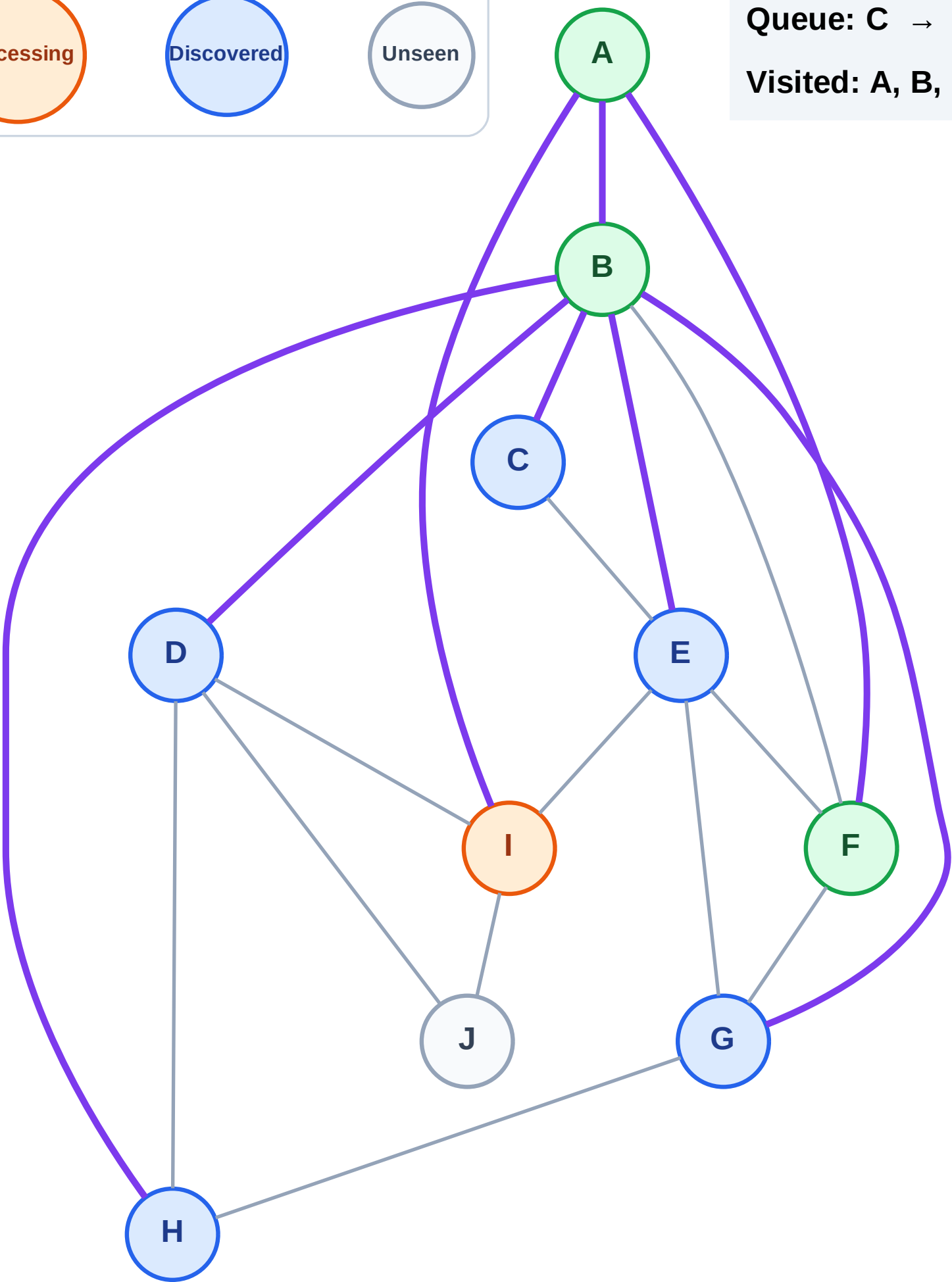
Processing

Discovered

Unseen

Queue: C → D → E → G → H

Visited: A, B, F



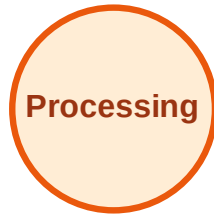
BFS Traversal

Step 9: Discover J from I

Legend



Complete



Processing



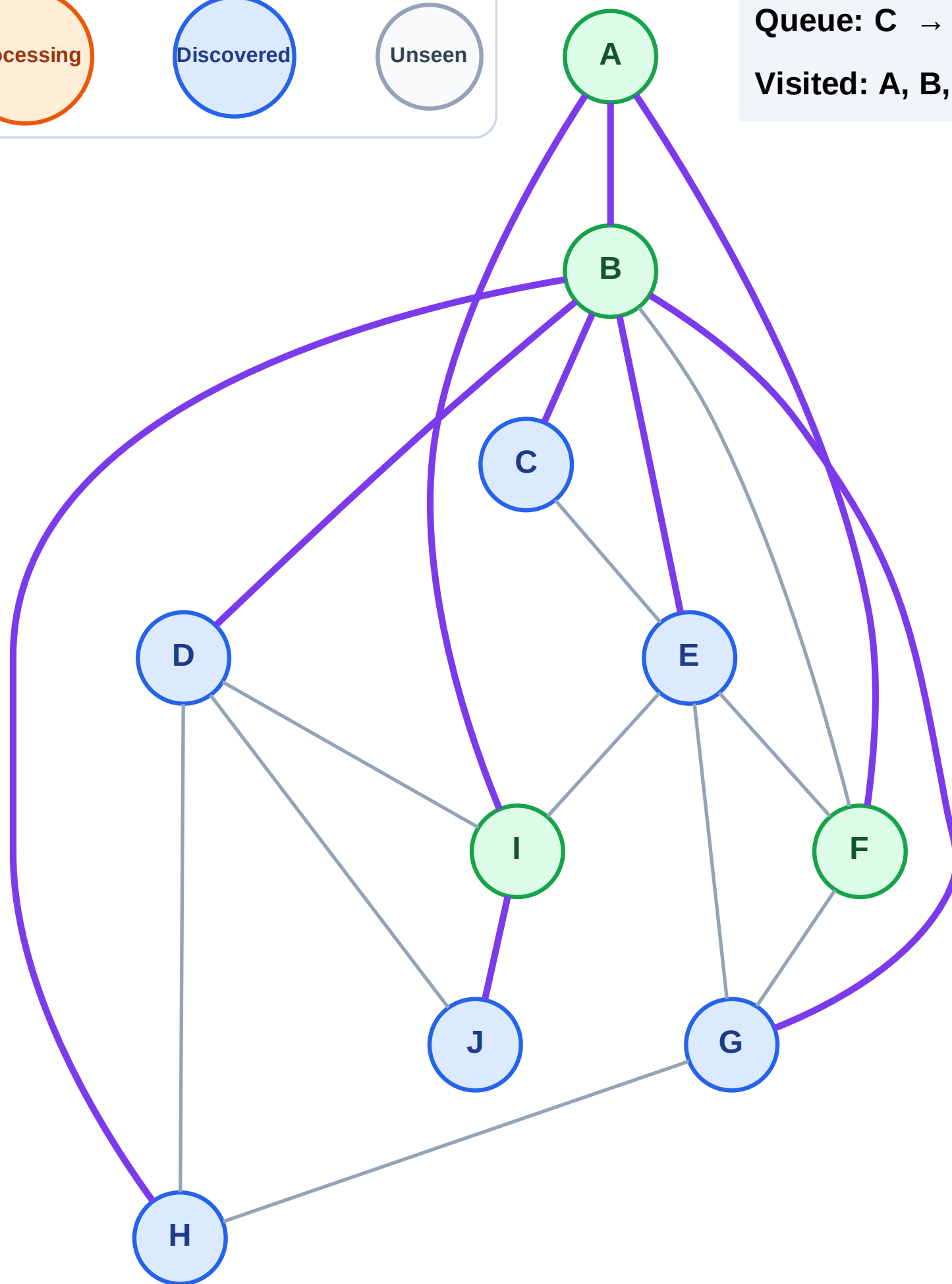
Discovered



Unseen

Queue: C → D → E → G → H → J

Visited: A, B, F, I



BFS Traversal

Step 10: Process node C

Legend



Complete



Processing



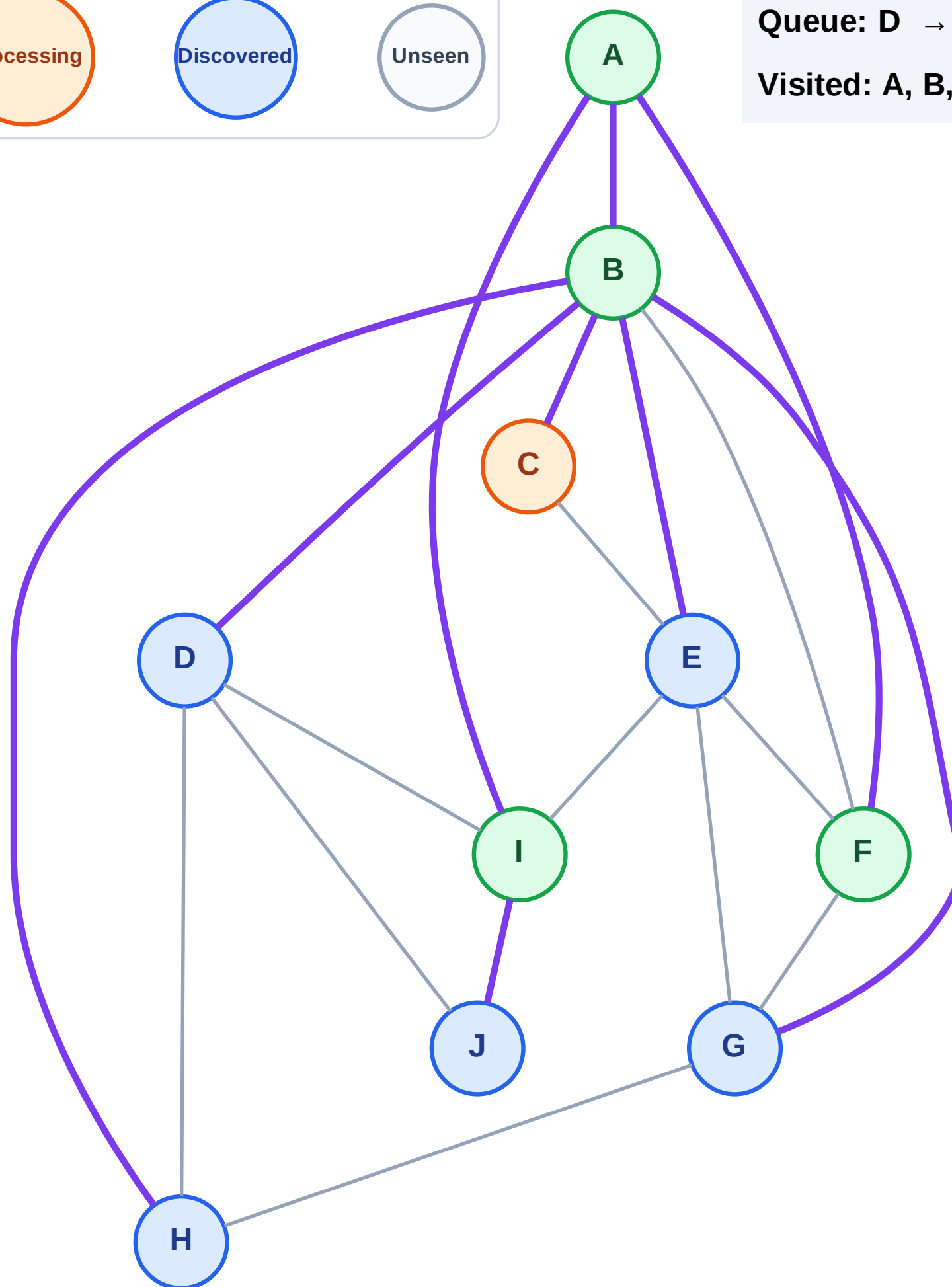
Discovered



Unseen

Queue: D → E → G → H → J

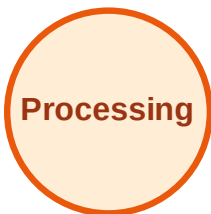
Visited: A, B, F, I



BFS Traversal

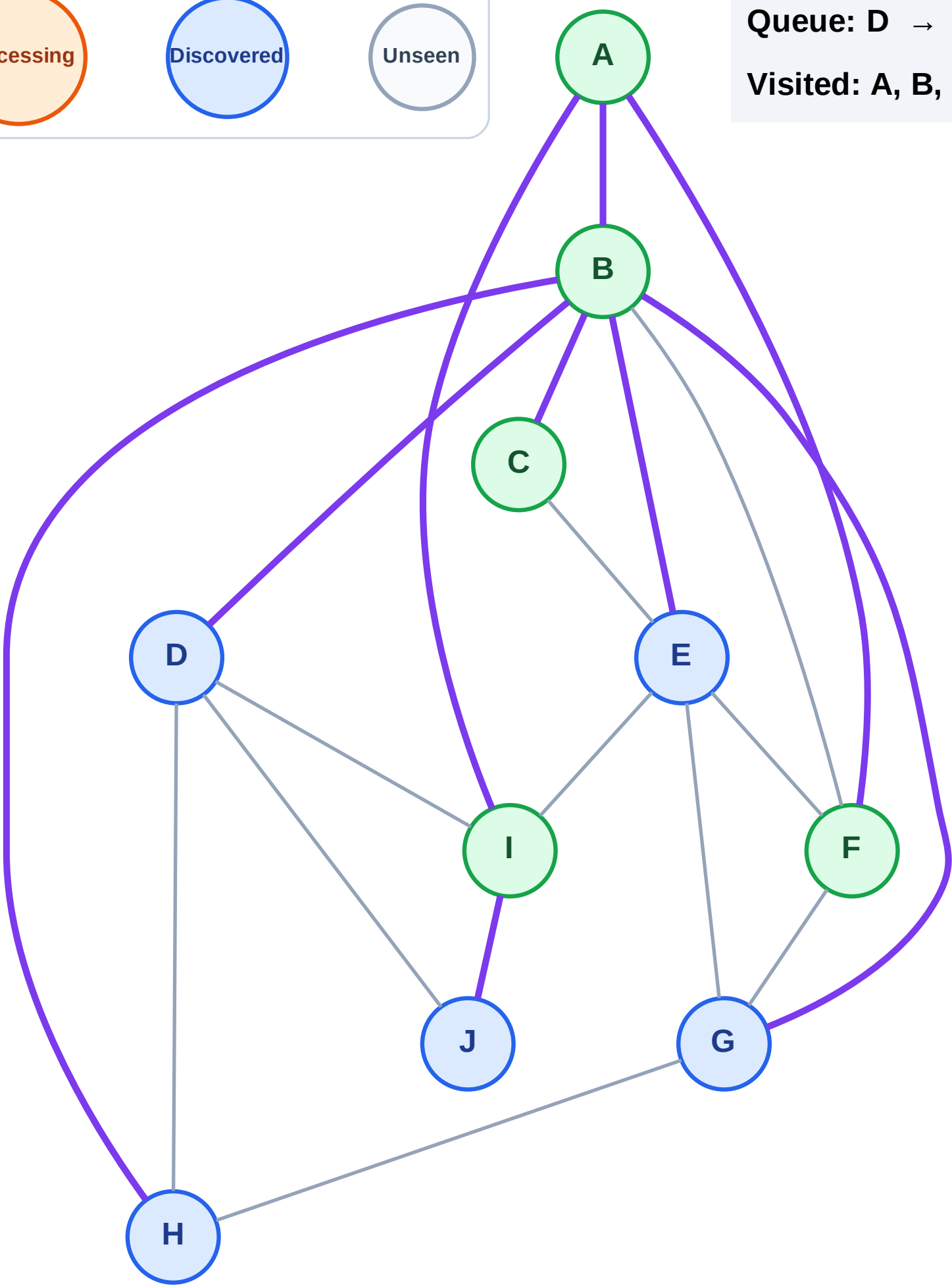
Step 11: No new neighbors from C

Legend



Queue: D → E → G → H → J

Visited: A, B, C, F, I



BFS Traversal

Step 12: Process node D

Legend

Complete

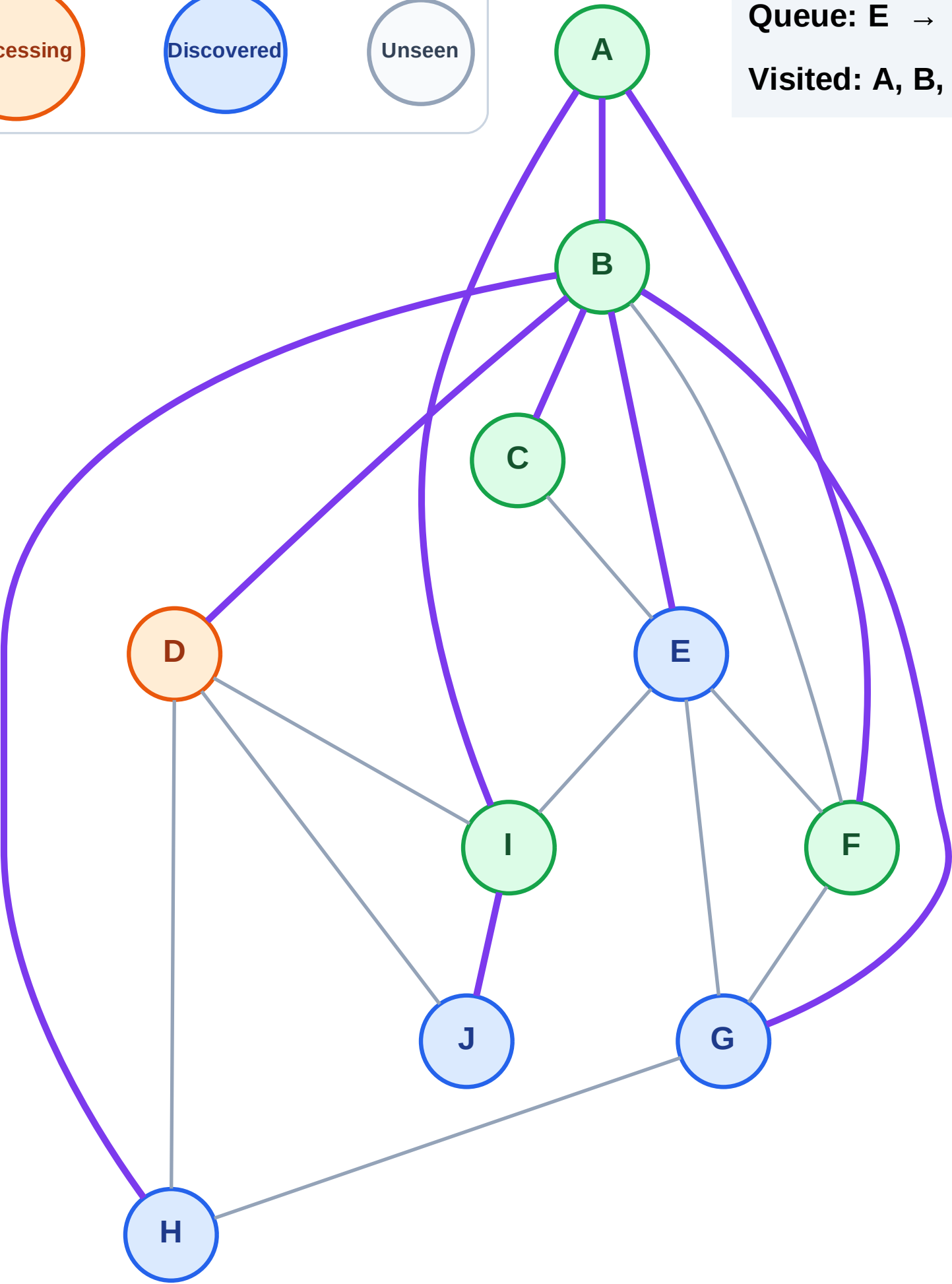
Processing

Discovered

Unseen

Queue: E → G → H → J

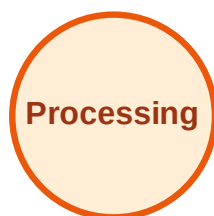
Visited: A, B, C, F, I



BFS Traversal

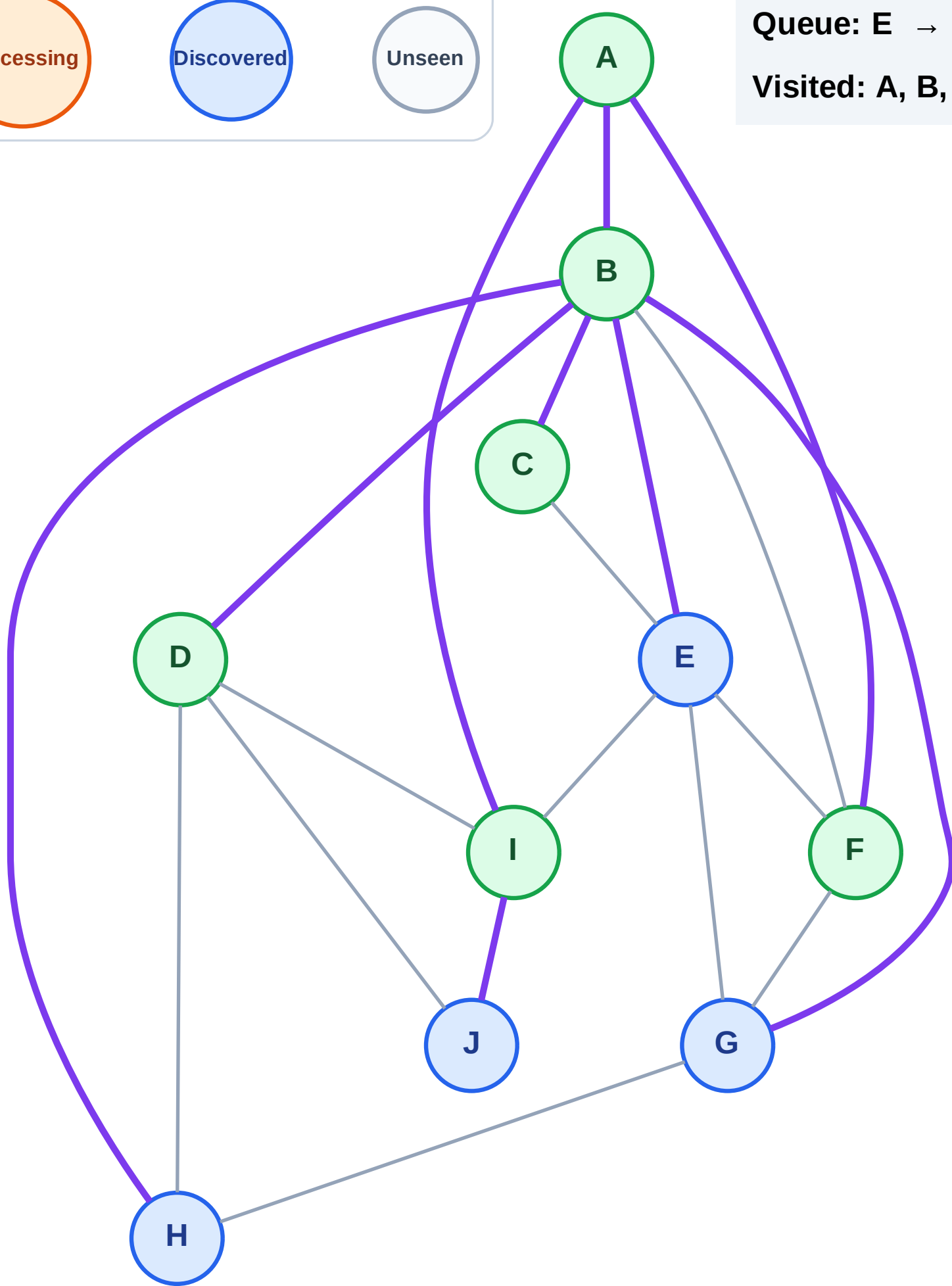
Step 13: No new neighbors from D

Legend



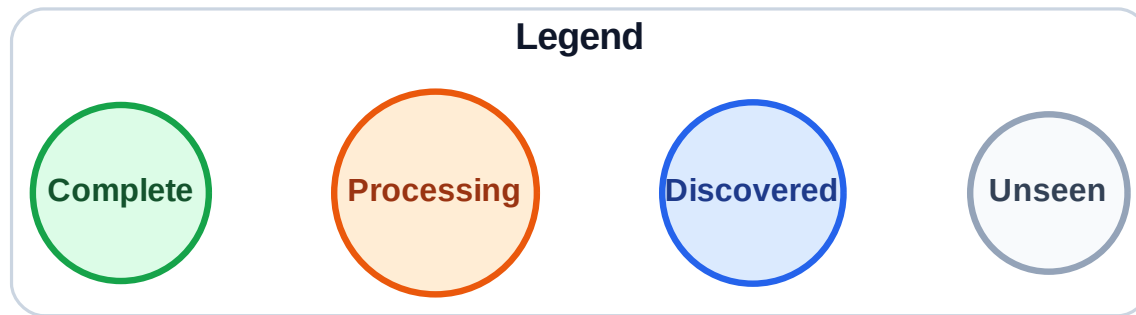
Queue: E → G → H → J

Visited: A, B, C, D, F, I

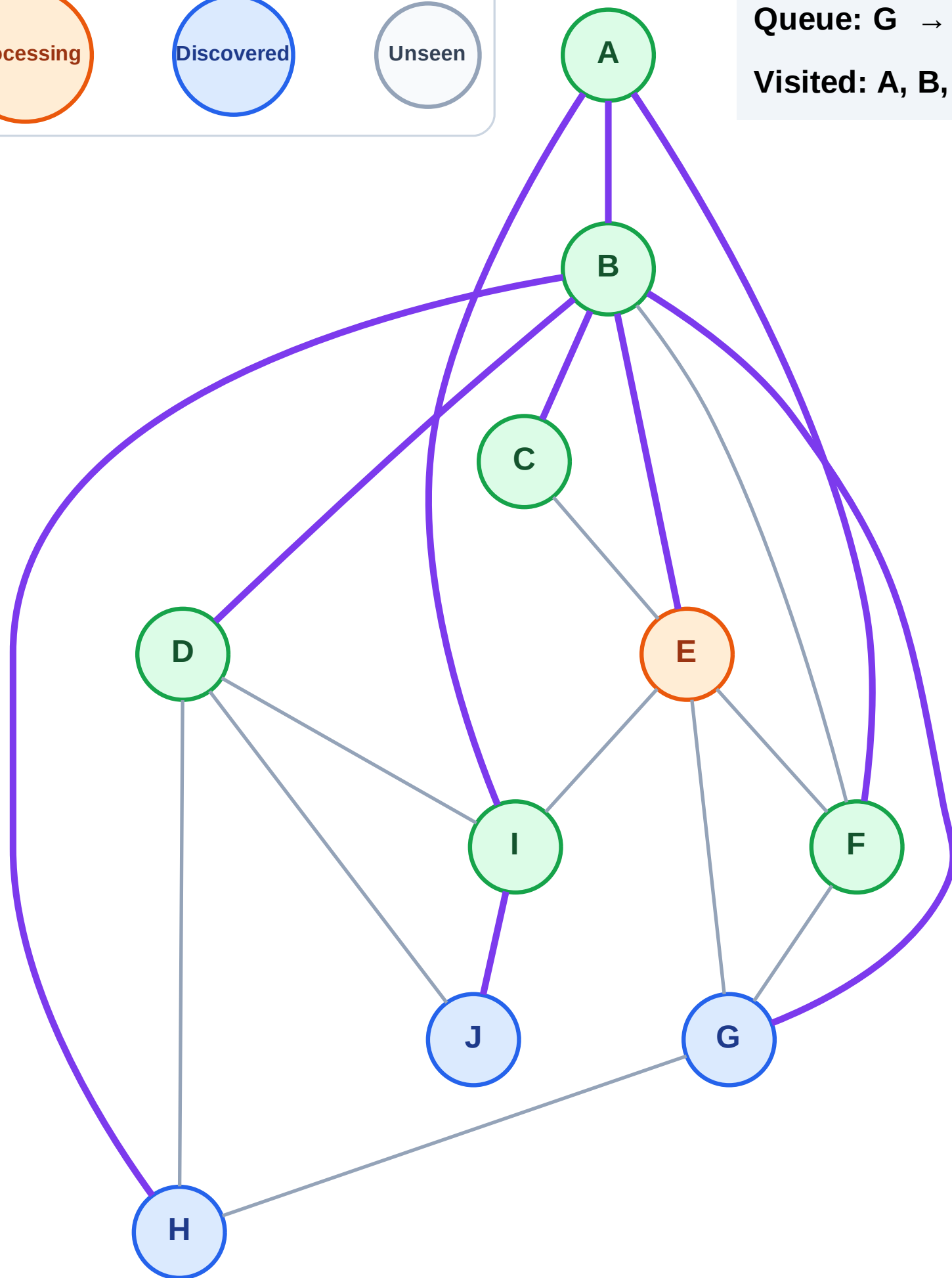


BFS Traversal

Step 14: Process node E



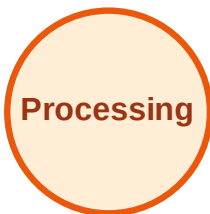
Queue: G → H → J
Visited: A, B, C, D, F, I



BFS Traversal

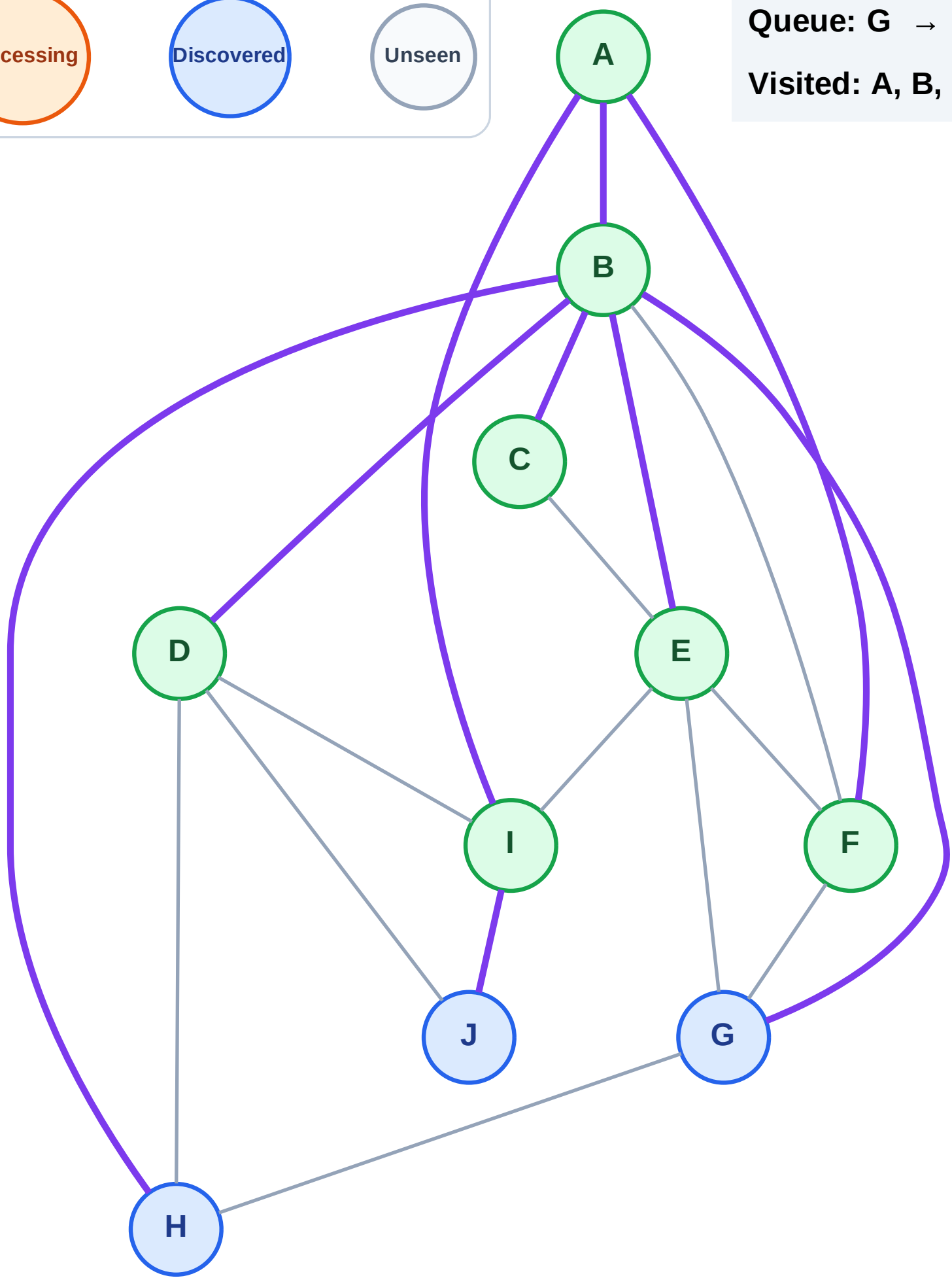
Step 15: No new neighbors from E

Legend



Queue: G → H → J

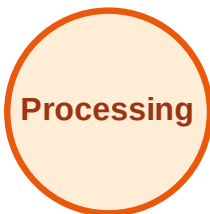
Visited: A, B, C, D, E, F, I



BFS Traversal

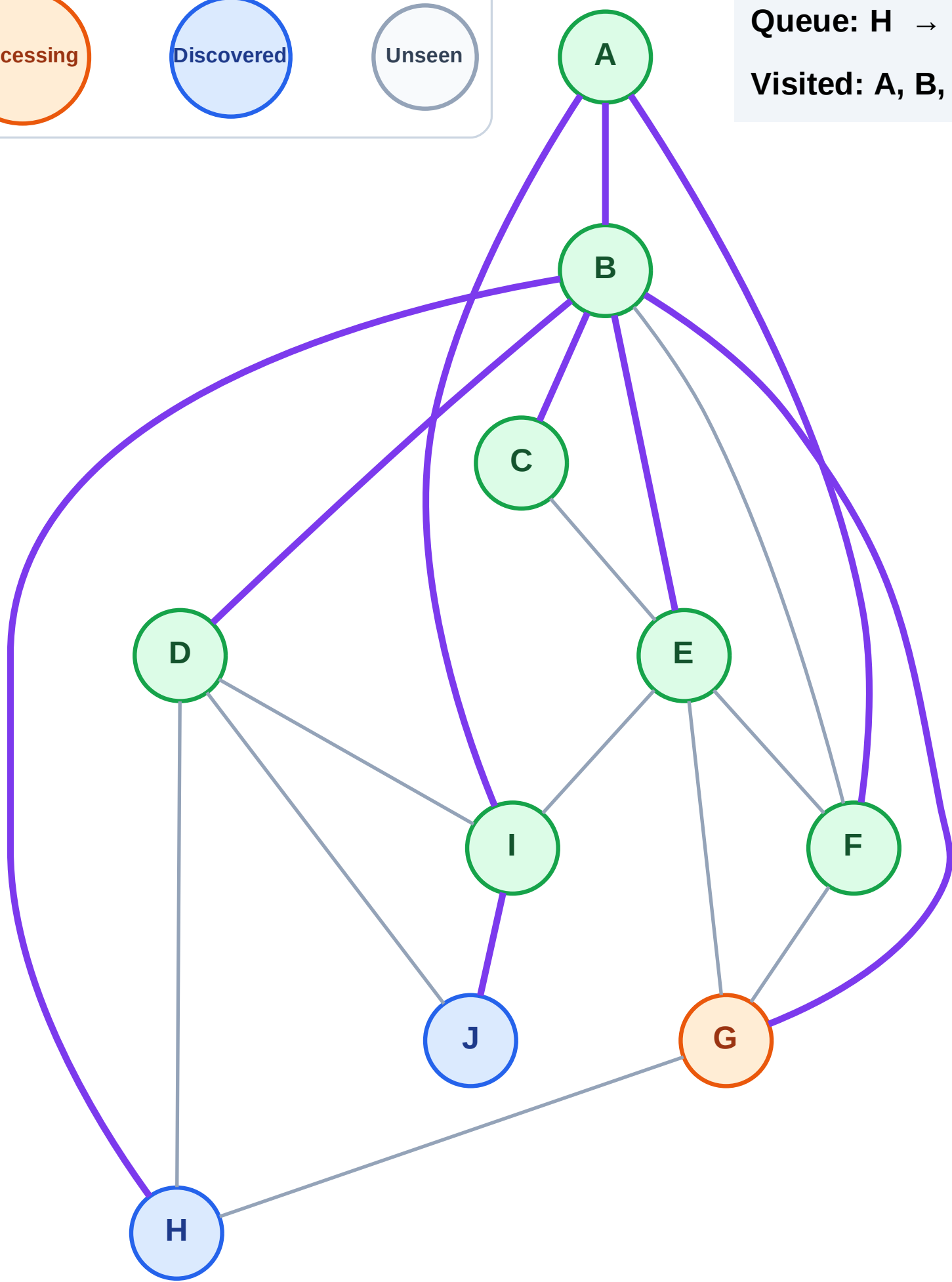
Step 16: Process node G

Legend



Queue: H → J

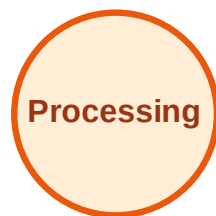
Visited: A, B, C, D, E, F, I



BFS Traversal

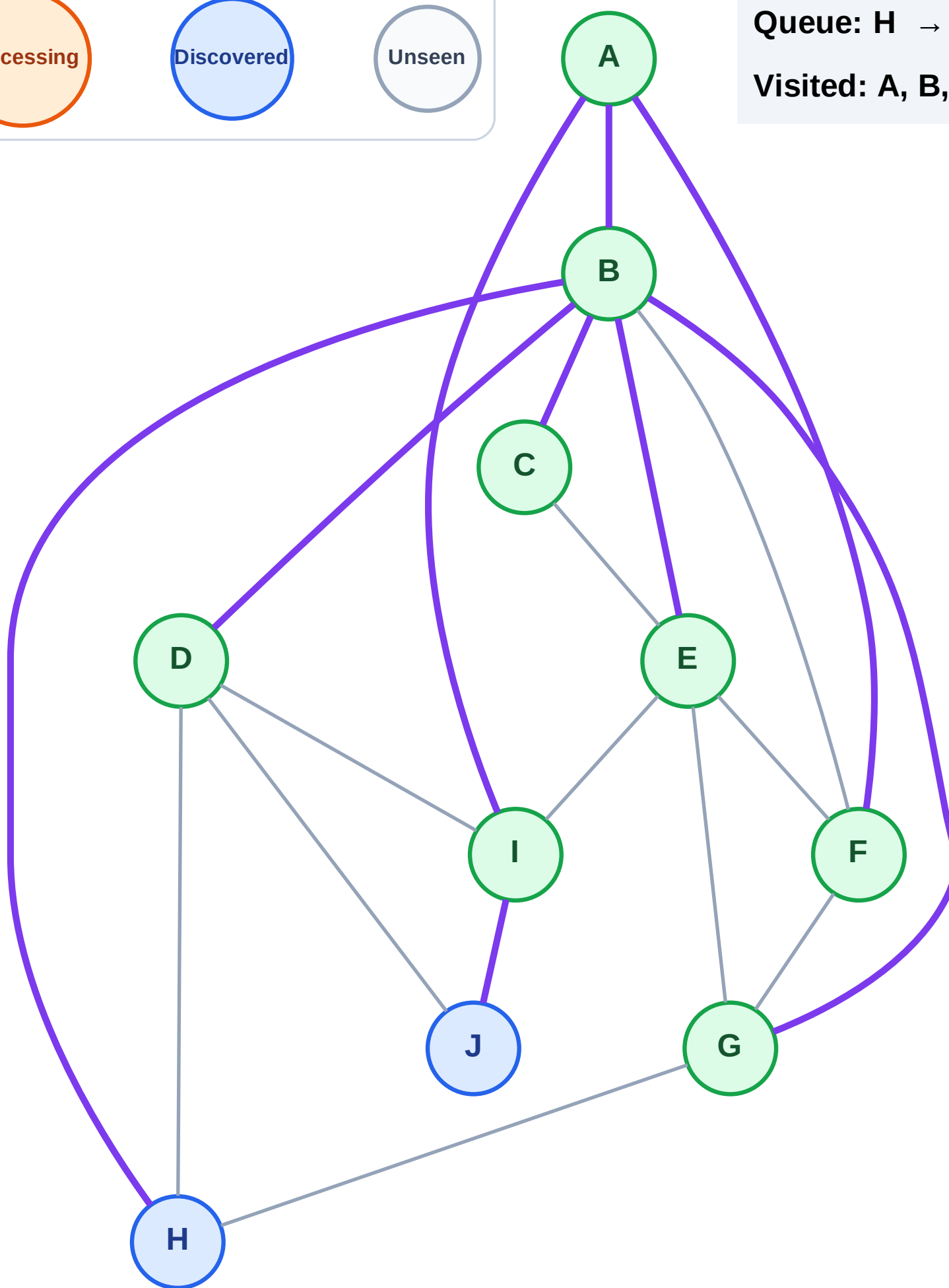
Step 17: No new neighbors from G

Legend



Queue: H → J

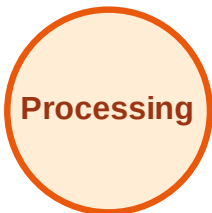
Visited: A, B, C, D, E, F, G, I



BFS Traversal

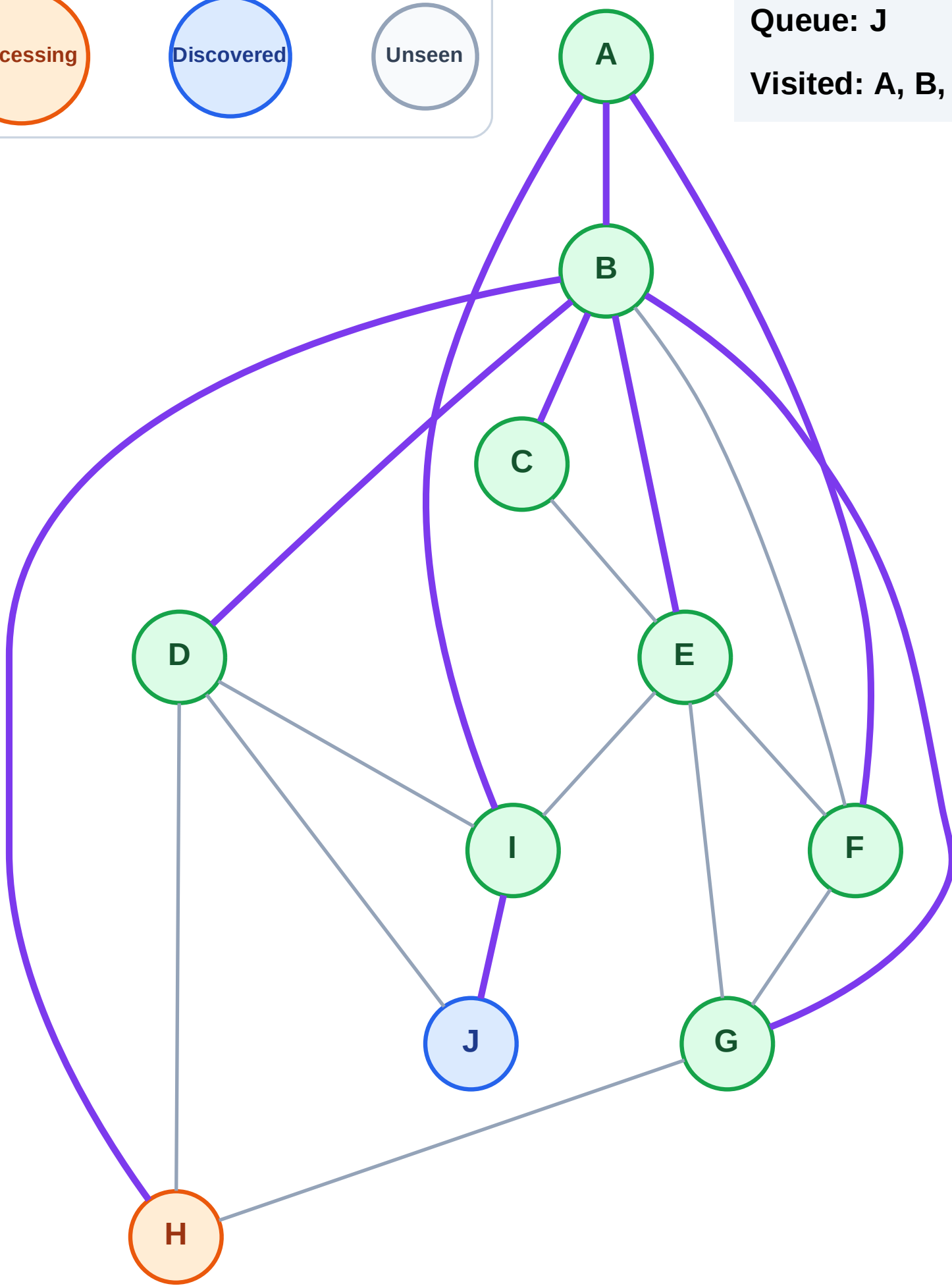
Step 18: Process node H

Legend



Queue: J

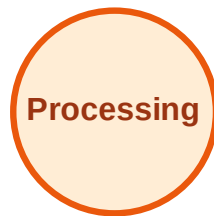
Visited: A, B, C, D, E, F, G, I



BFS Traversal

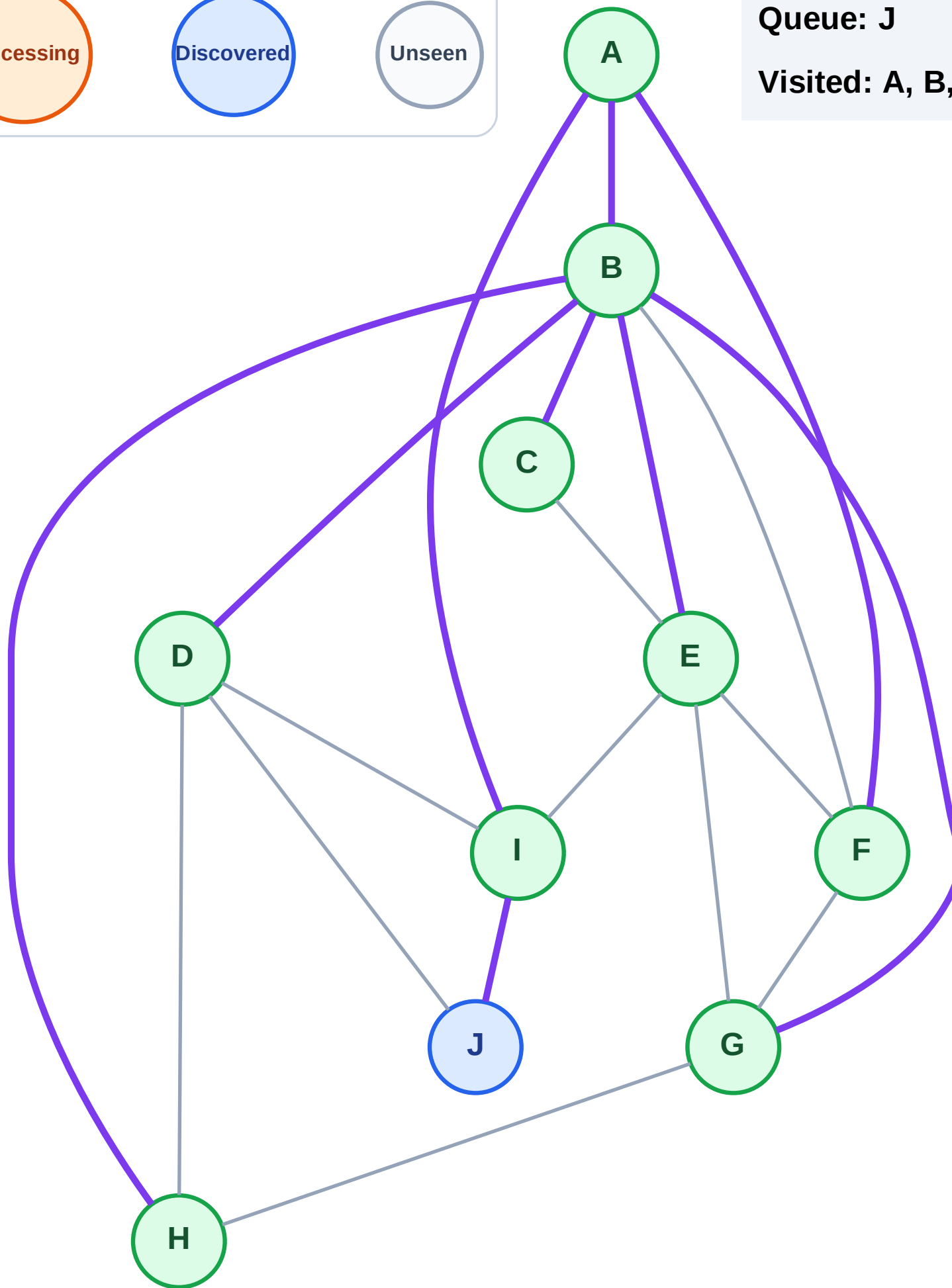
Step 19: No new neighbors from H

Legend



Queue: J

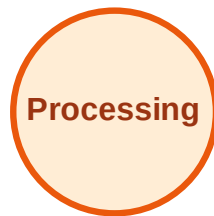
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

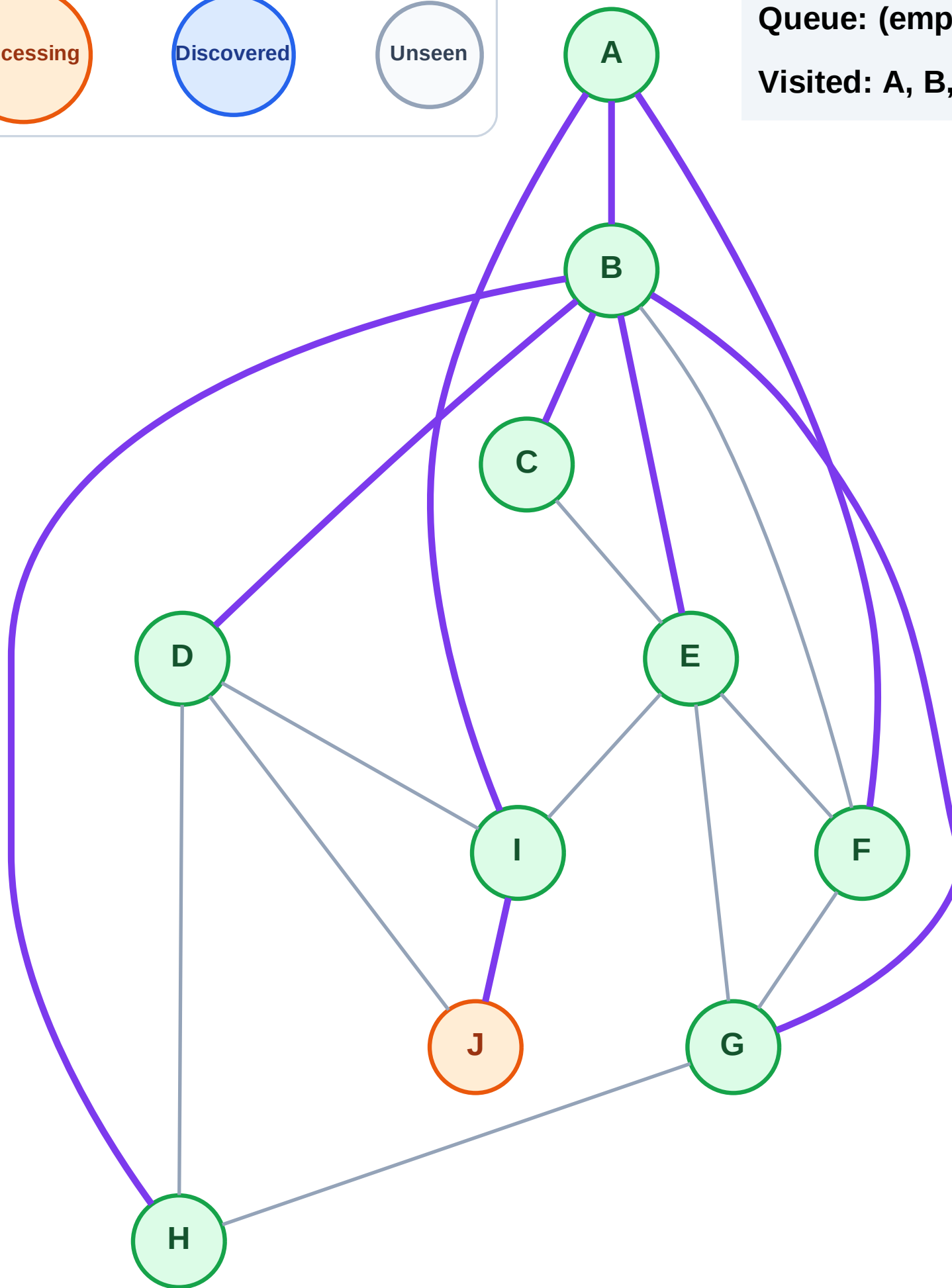
Step 20: Process node J

Legend



Queue: (empty)

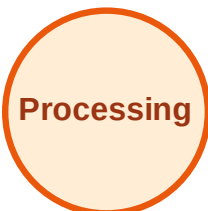
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

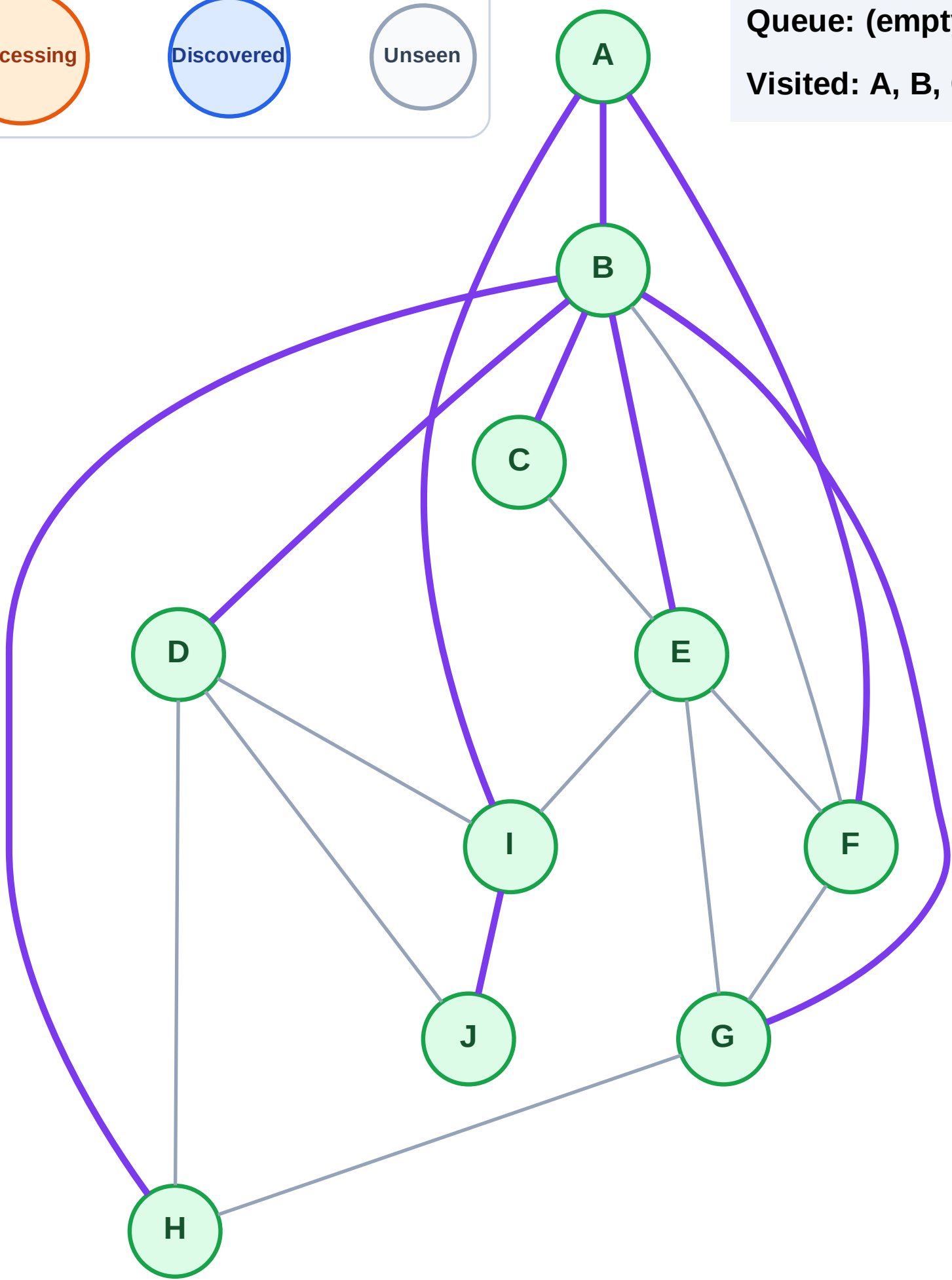
Step 21: No new neighbors from J

Legend



Queue: (empty)

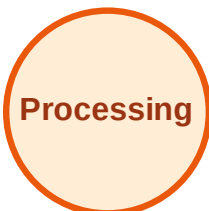
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

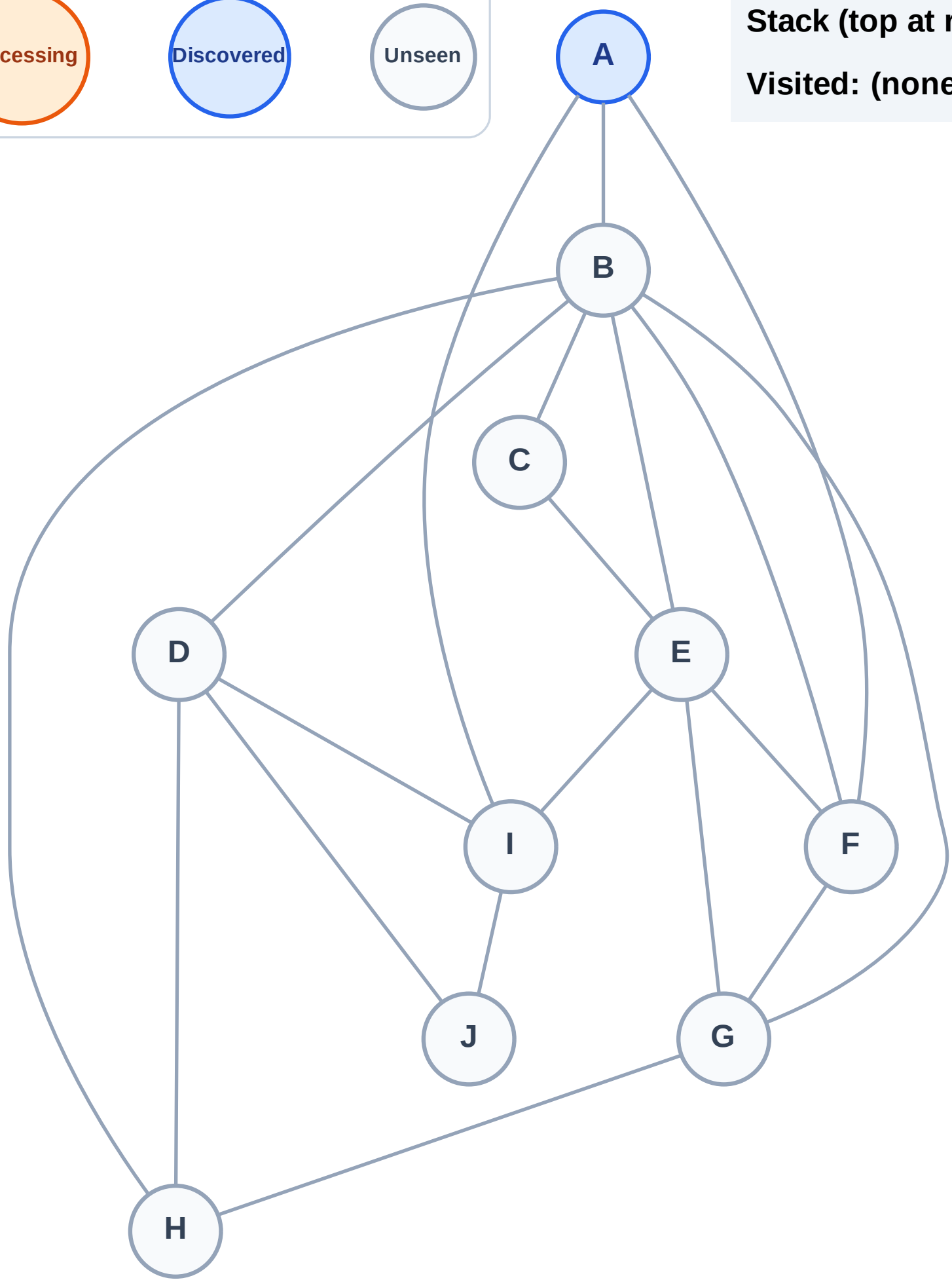
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

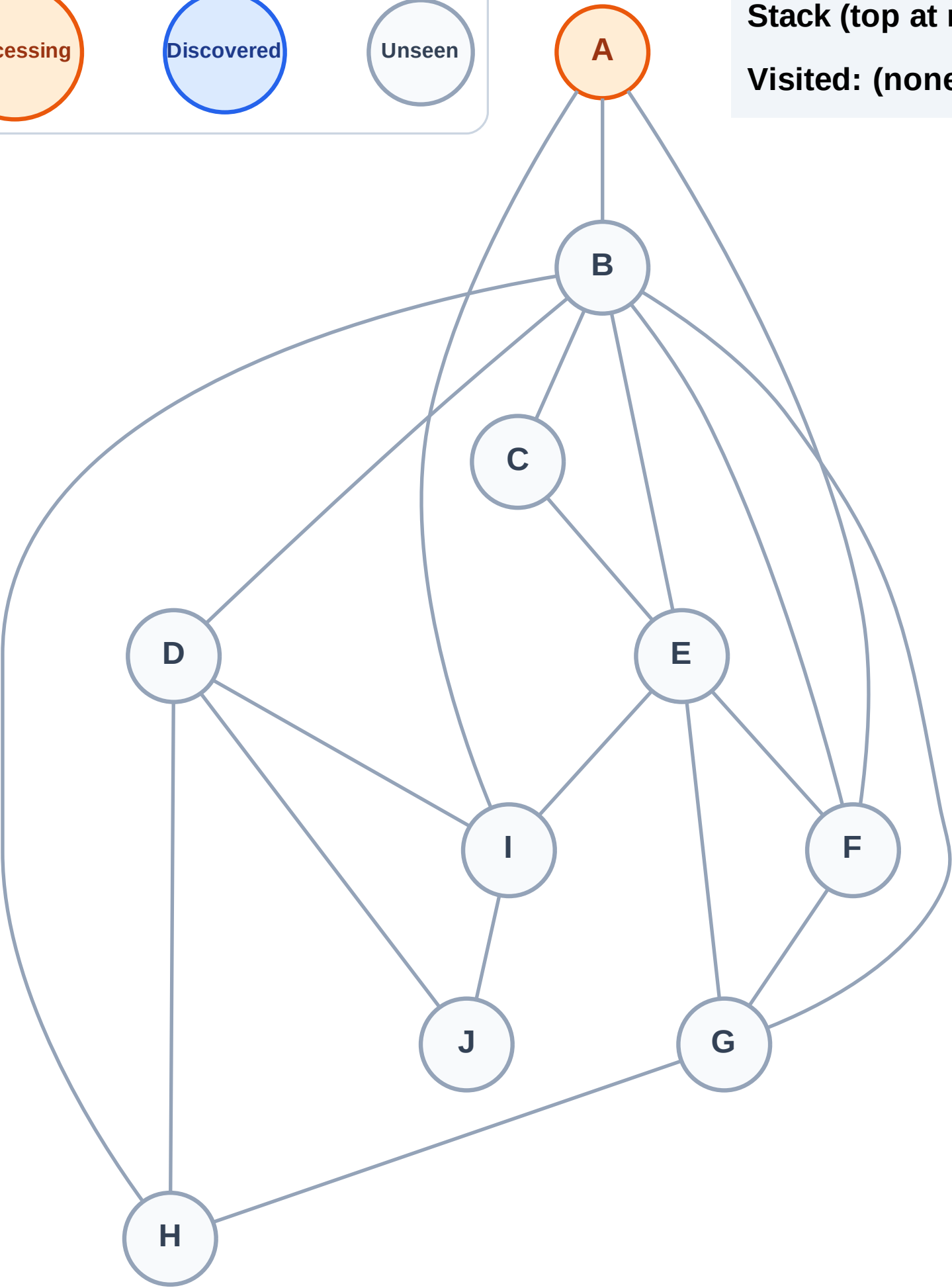
Complete

Processing

Discovered

Unseen

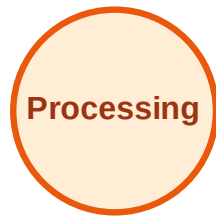
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

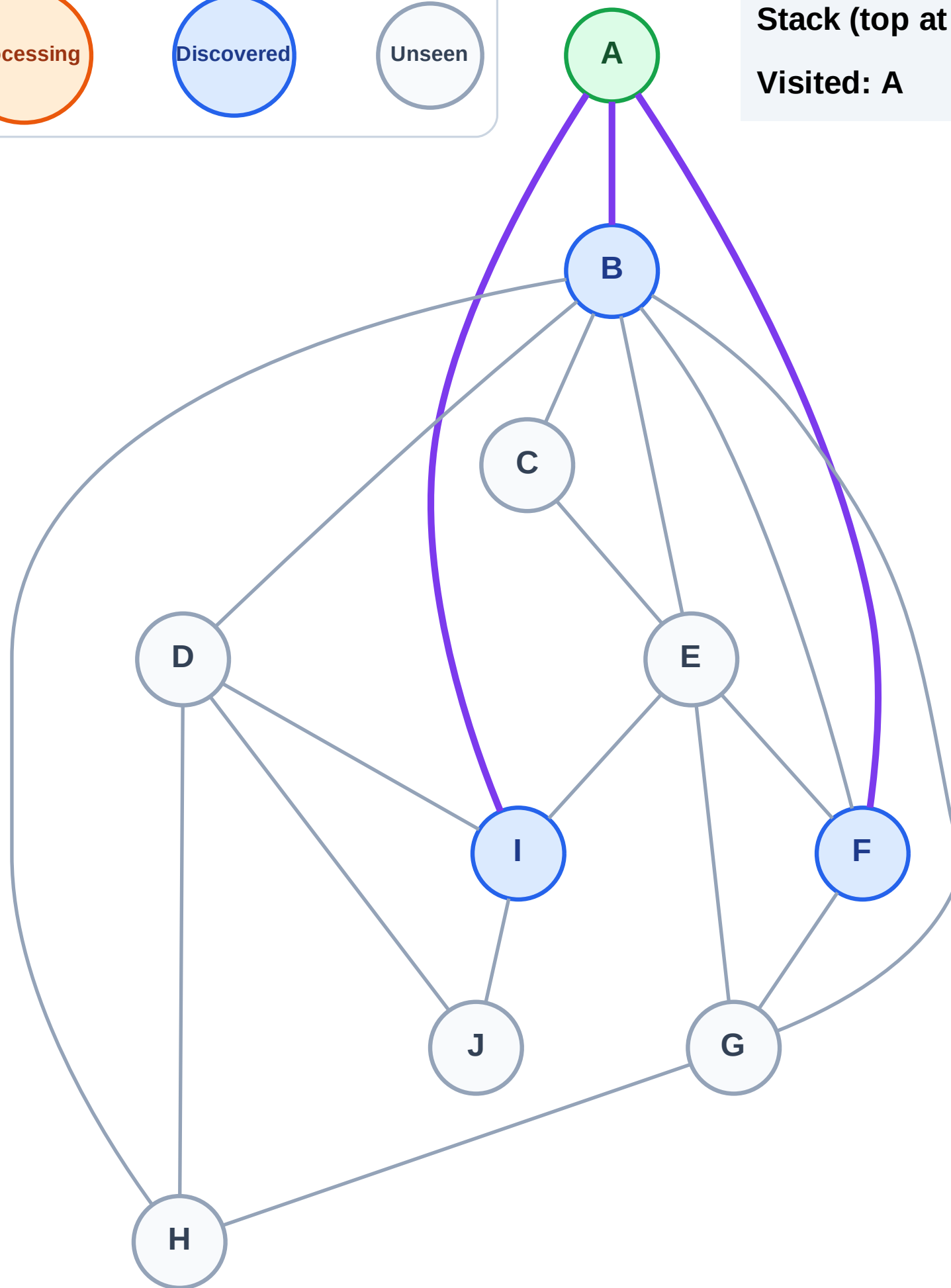
Step 3: Discover I, F, B from A

Legend



Stack (top at right): I | F | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

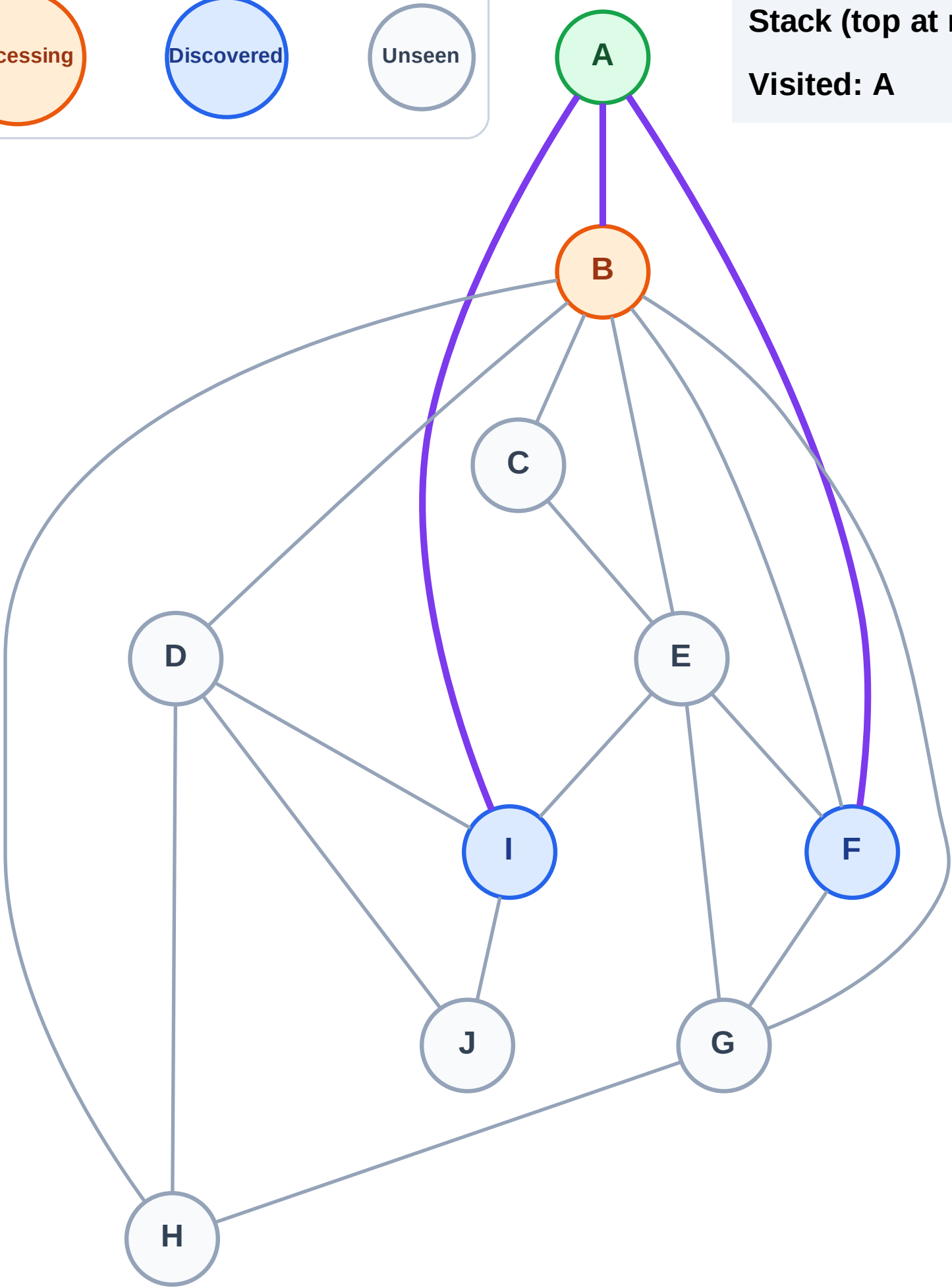
Processing

Discovered

Unseen

Stack (top at right): I | F

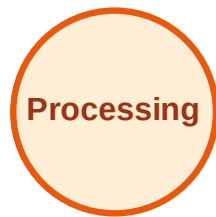
Visited: A



DFS Traversal

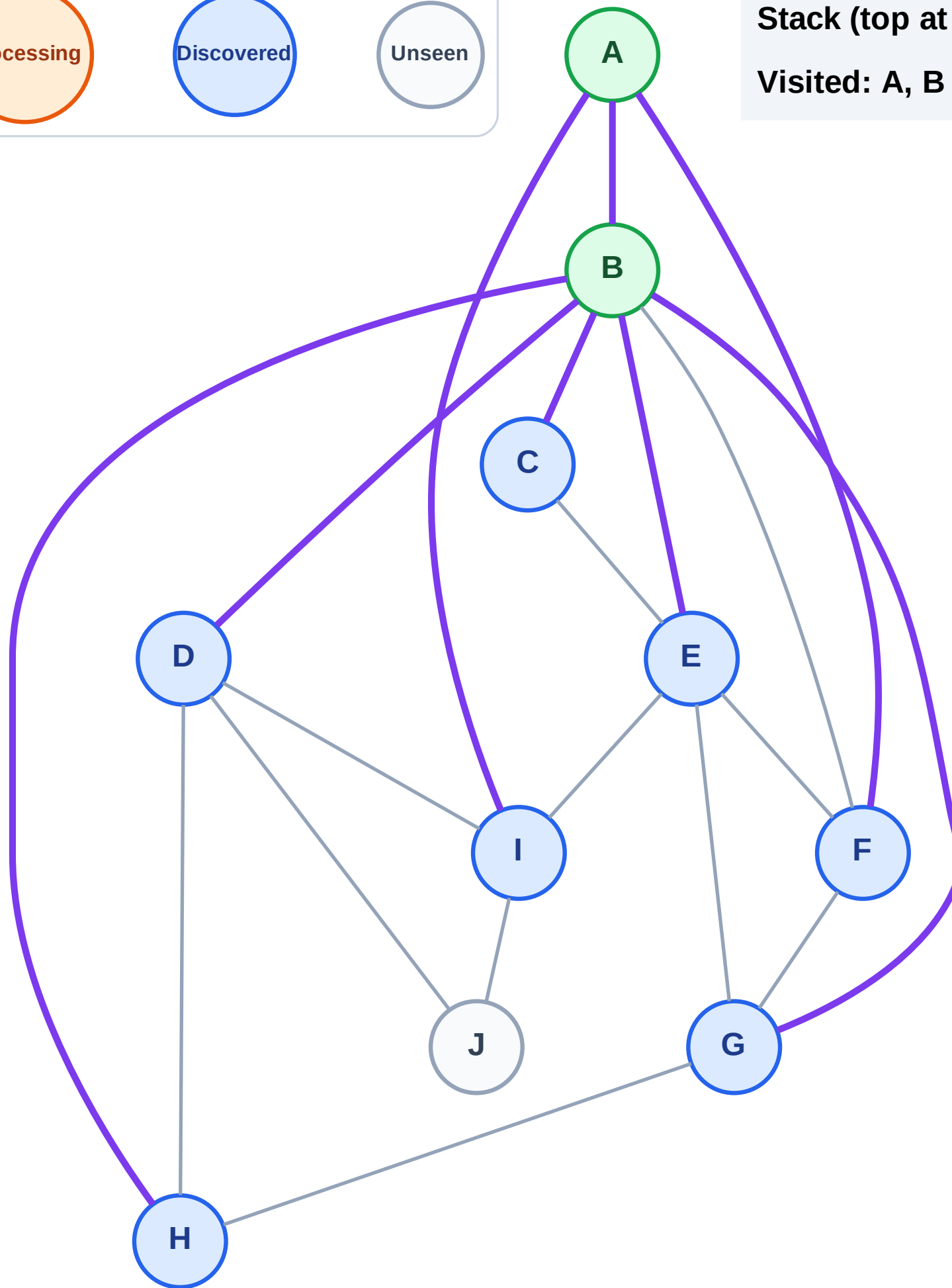
Step 5: Discover H, G, E, D, C from B

Legend



Stack (top at right): I | F | H | G | E | D | C

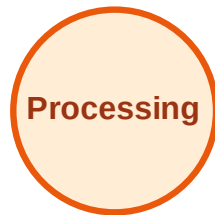
Visited: A, B



DFS Traversal

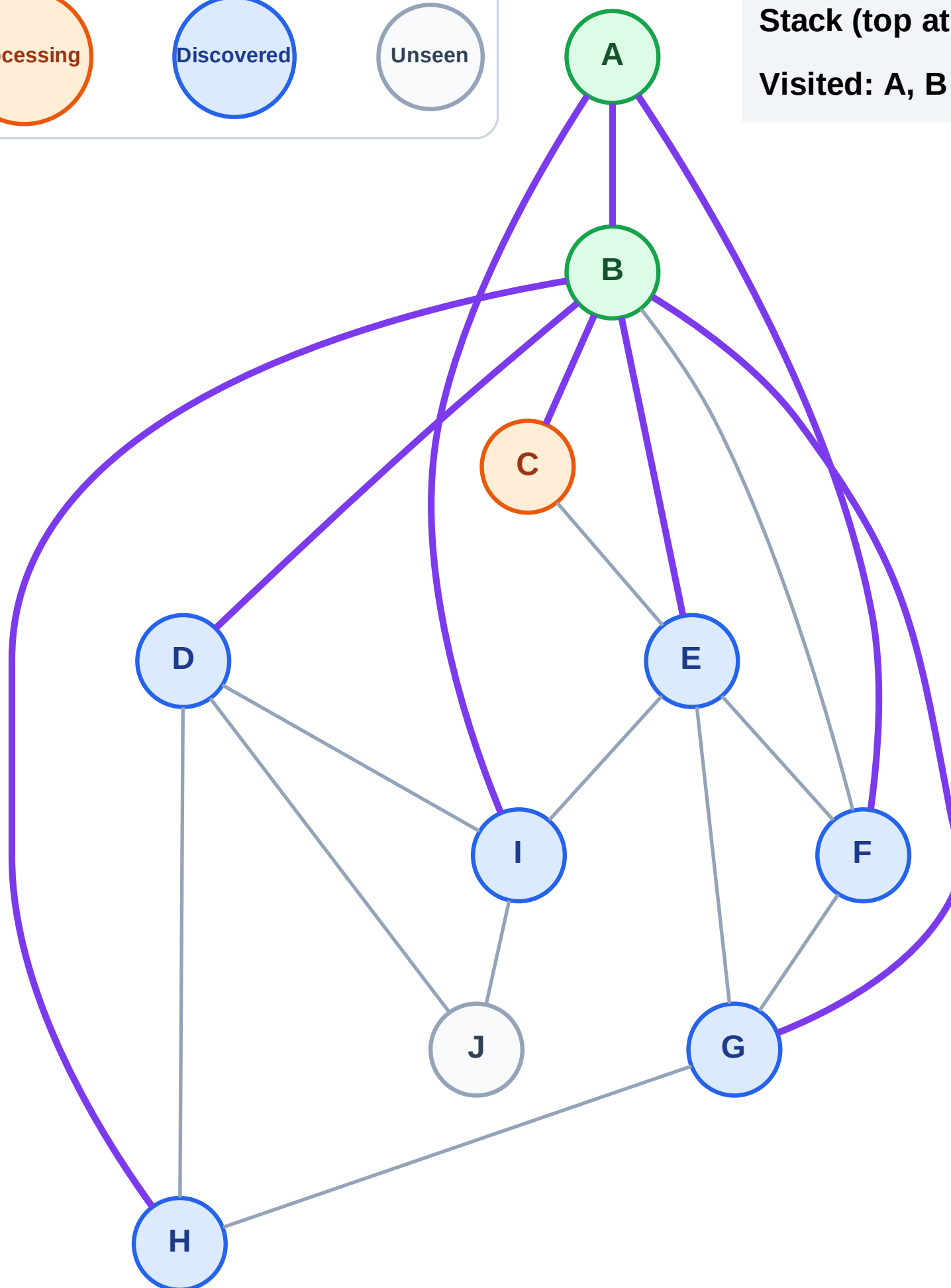
Step 6: Process node C

Legend



Stack (top at right): I | F | H | G | E | D

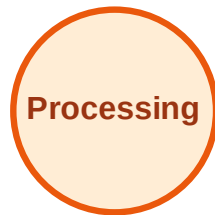
Visited: A, B



DFS Traversal

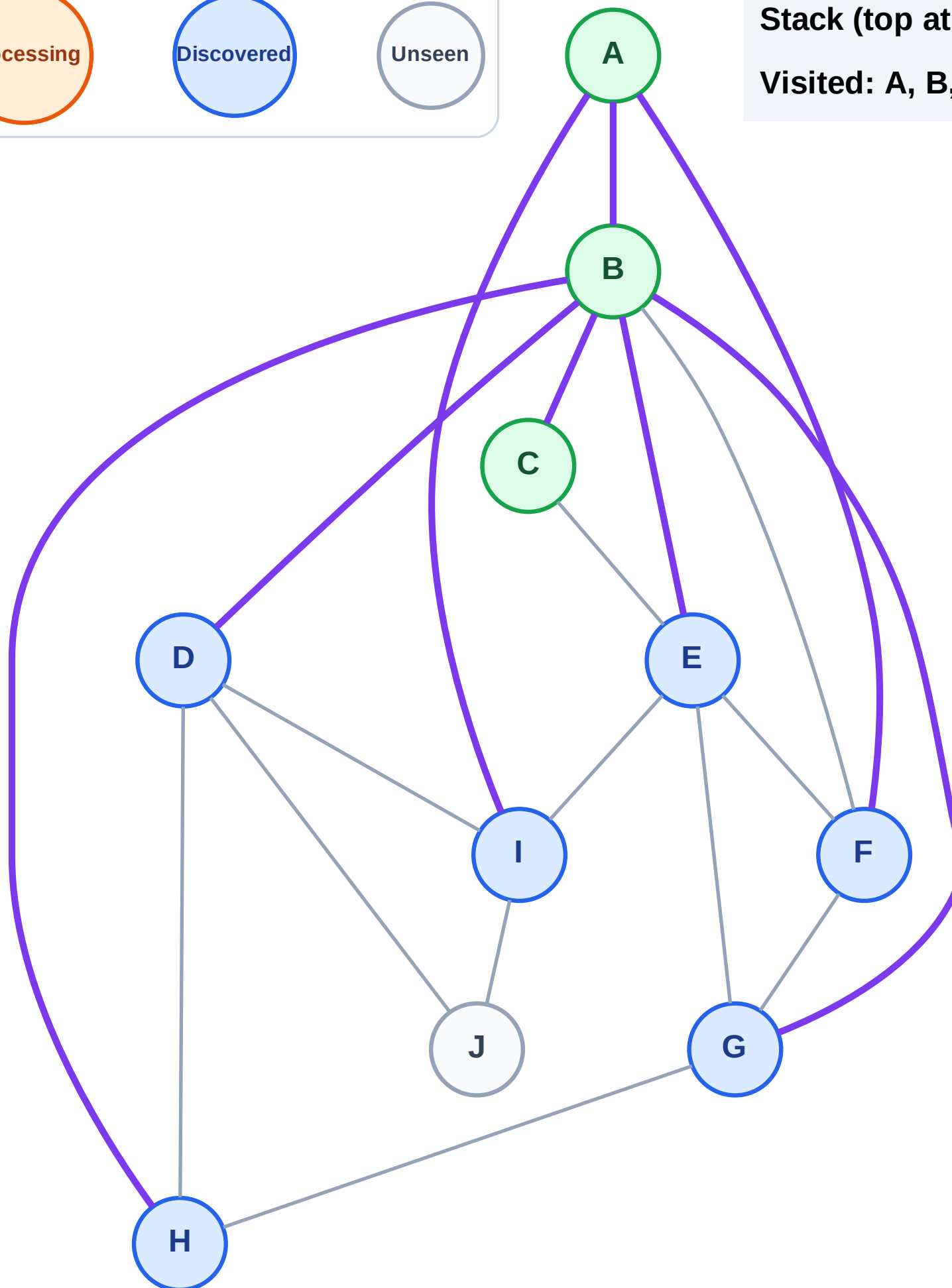
Step 7: No new neighbors from C

Legend



Stack (top at right): I | F | H | G | E | D

Visited: A, B, C



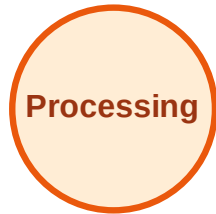
DFS Traversal

Step 8: Process node D

Legend



Complete



Processing



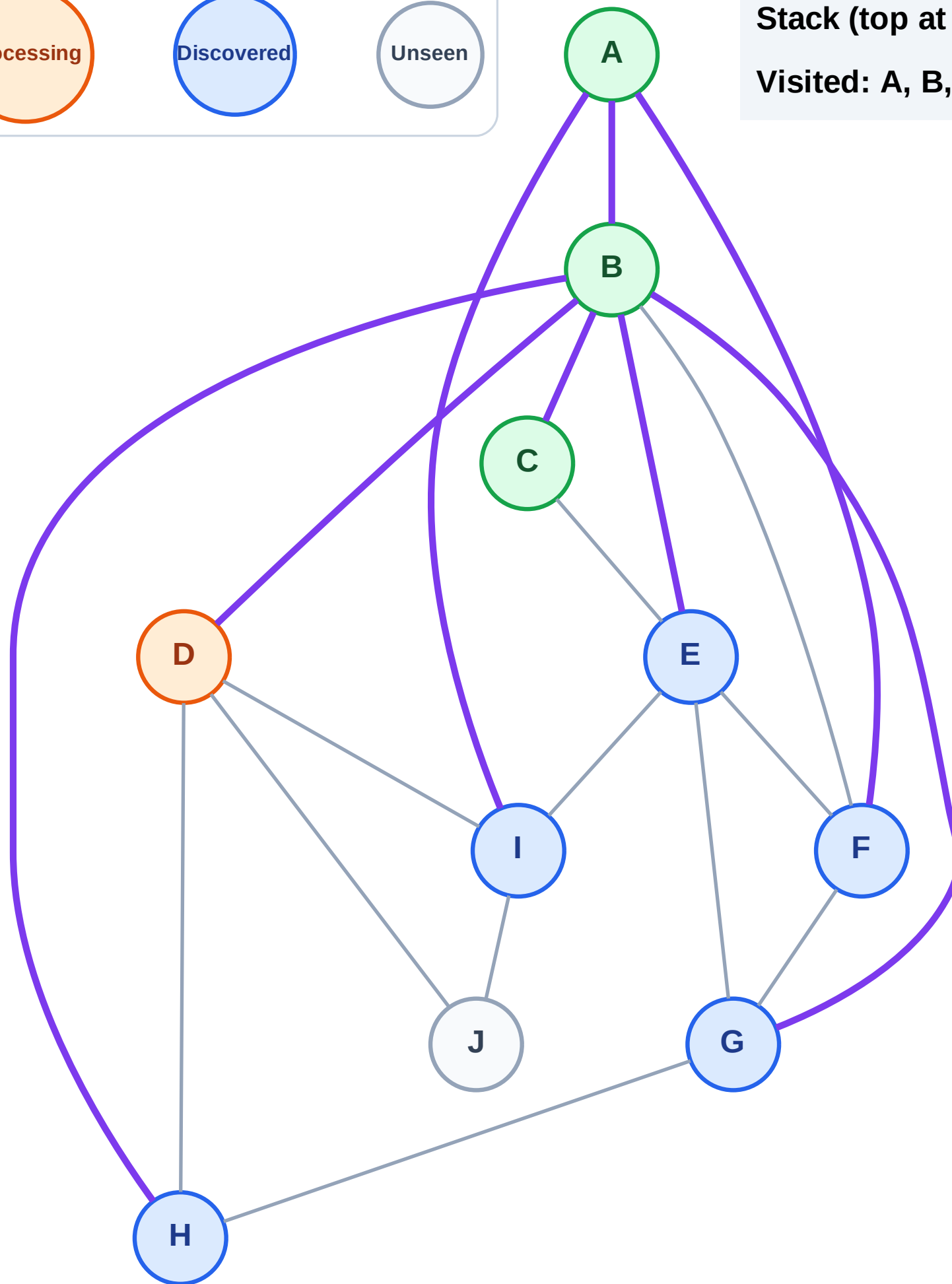
Discovered



Unseen

Stack (top at right): I | F | H | G | E

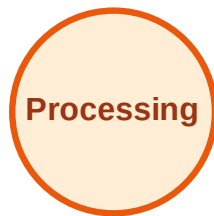
Visited: A, B, C



DFS Traversal

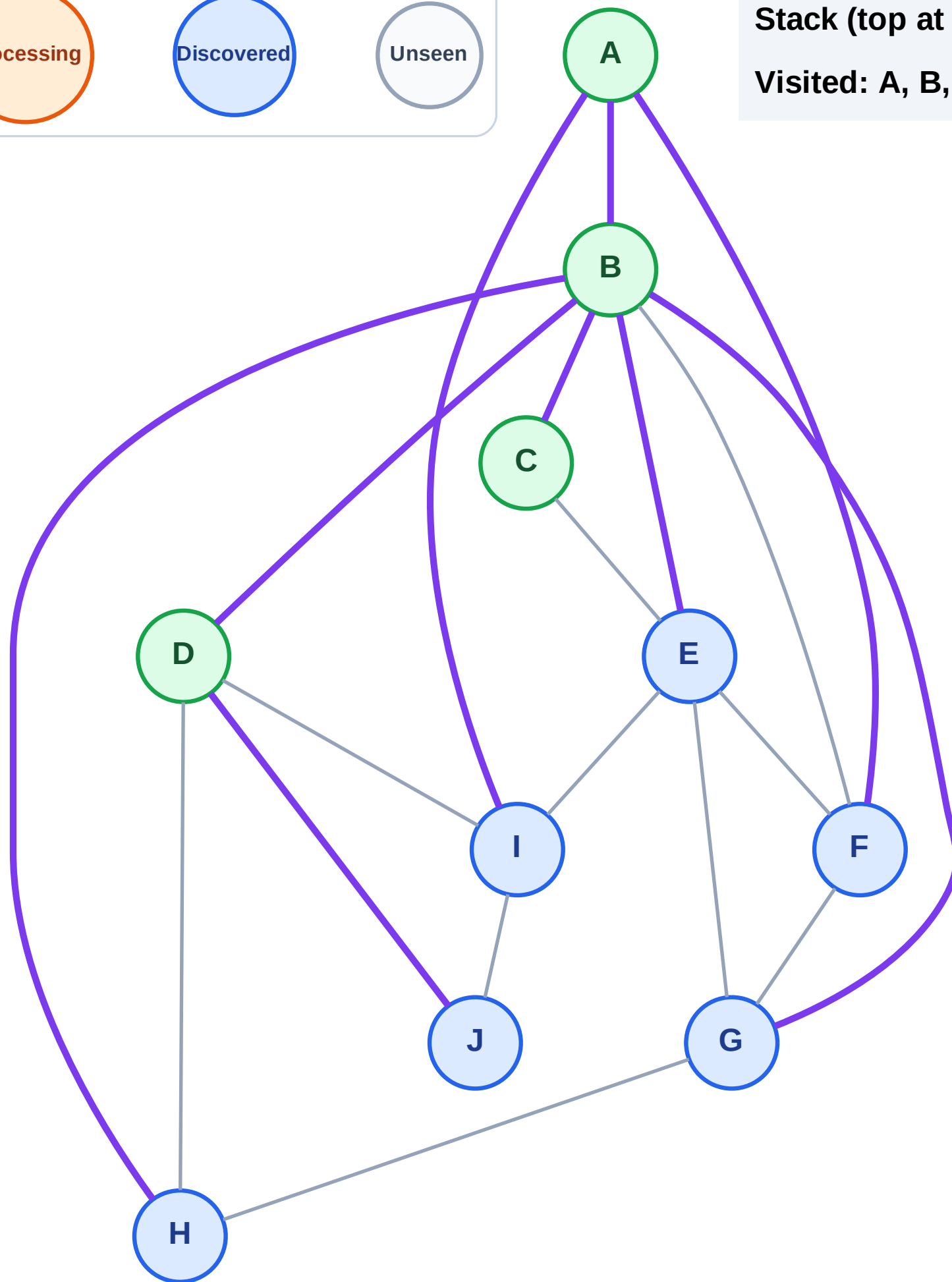
Step 9: Discover J from D

Legend

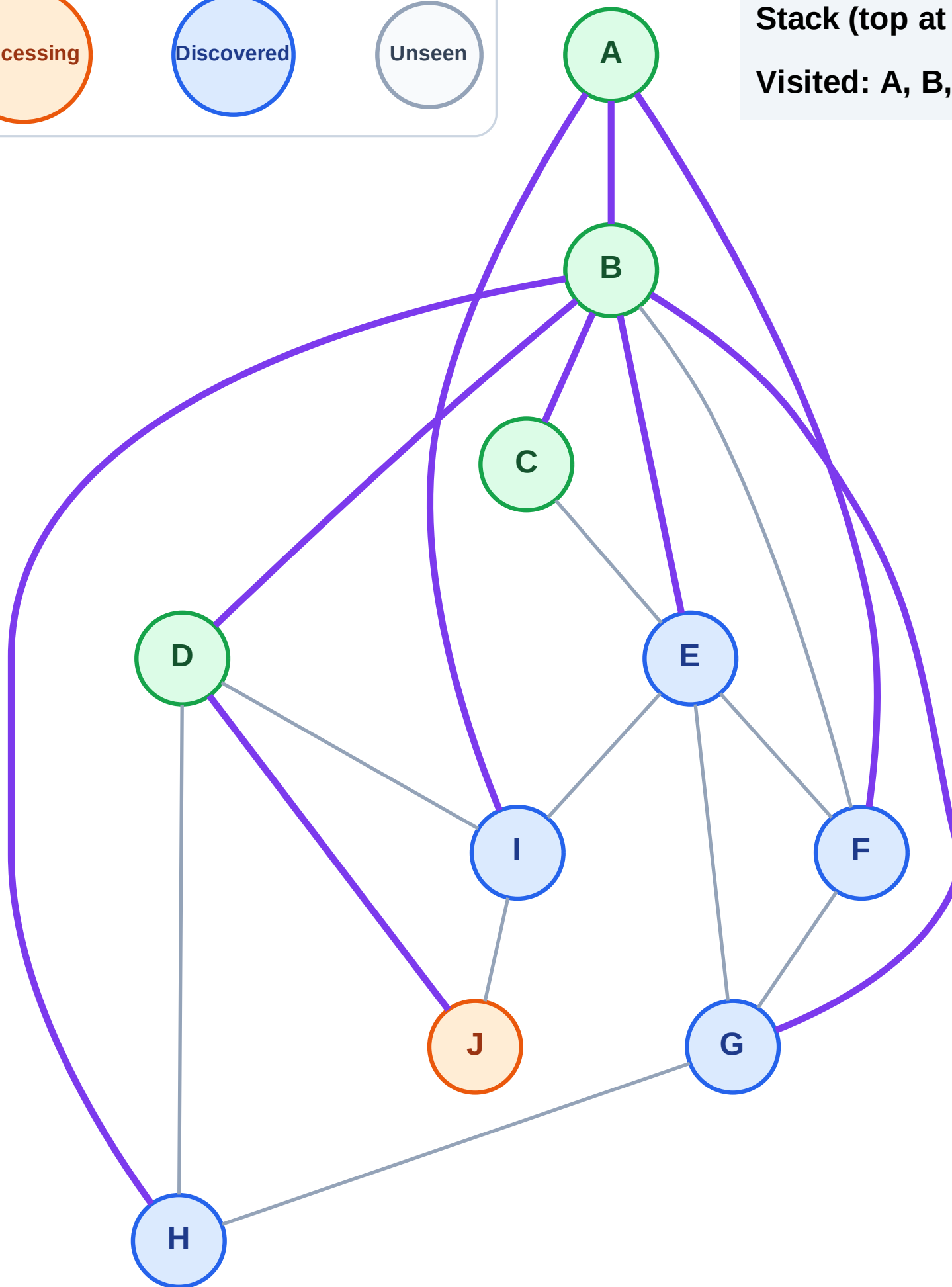
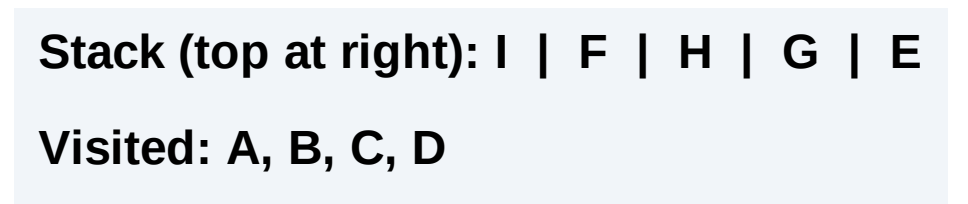


Stack (top at right): I | F | H | G | E | J

Visited: A, B, C, D



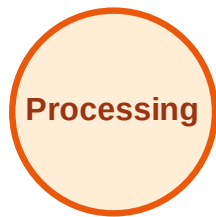
Step 10: Process node J



DFS Traversal

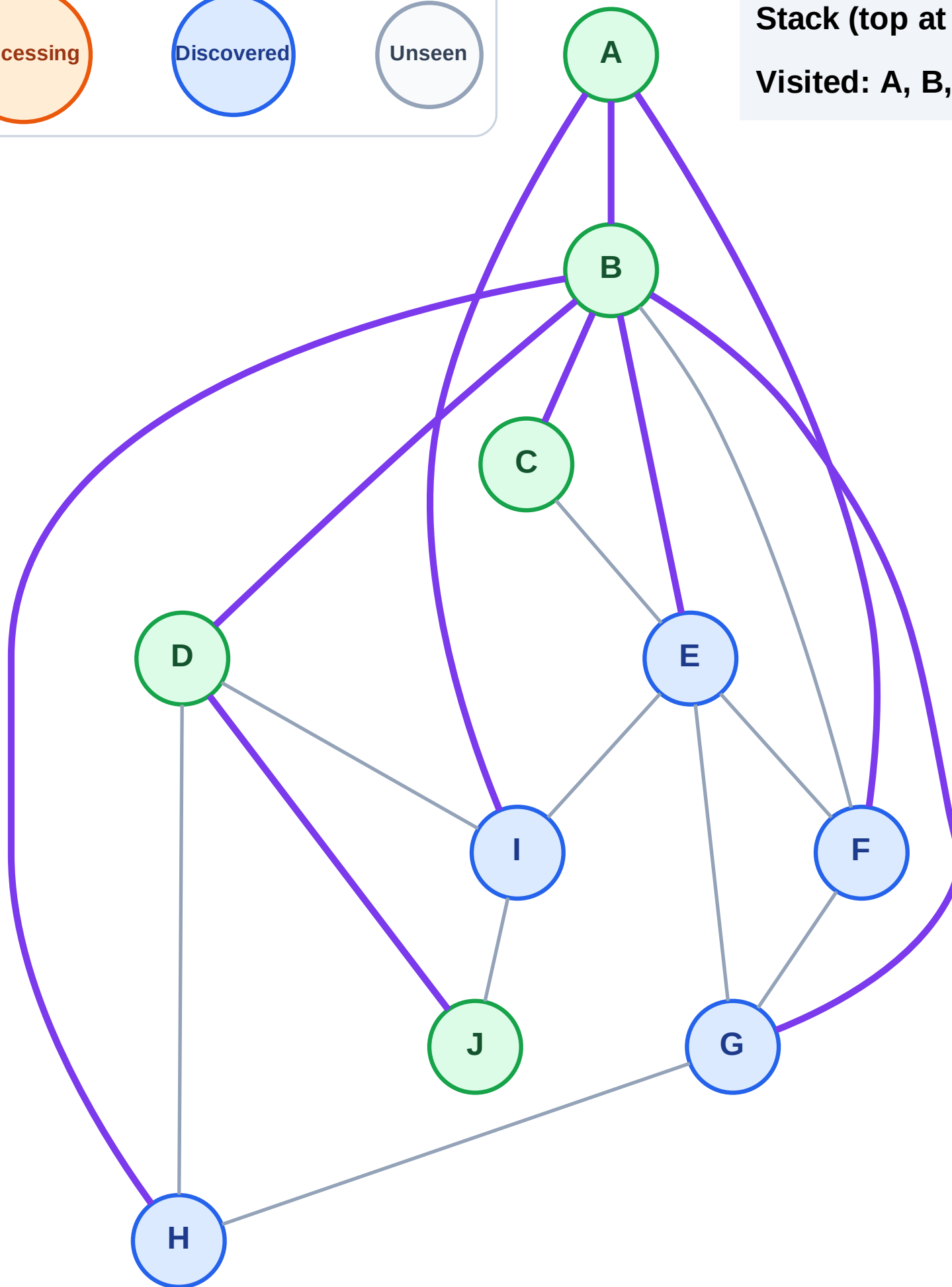
Step 11: No new neighbors from J

Legend



Stack (top at right): I | F | H | G | E

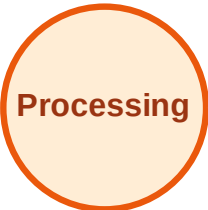
Visited: A, B, C, D, J



DFS Traversal

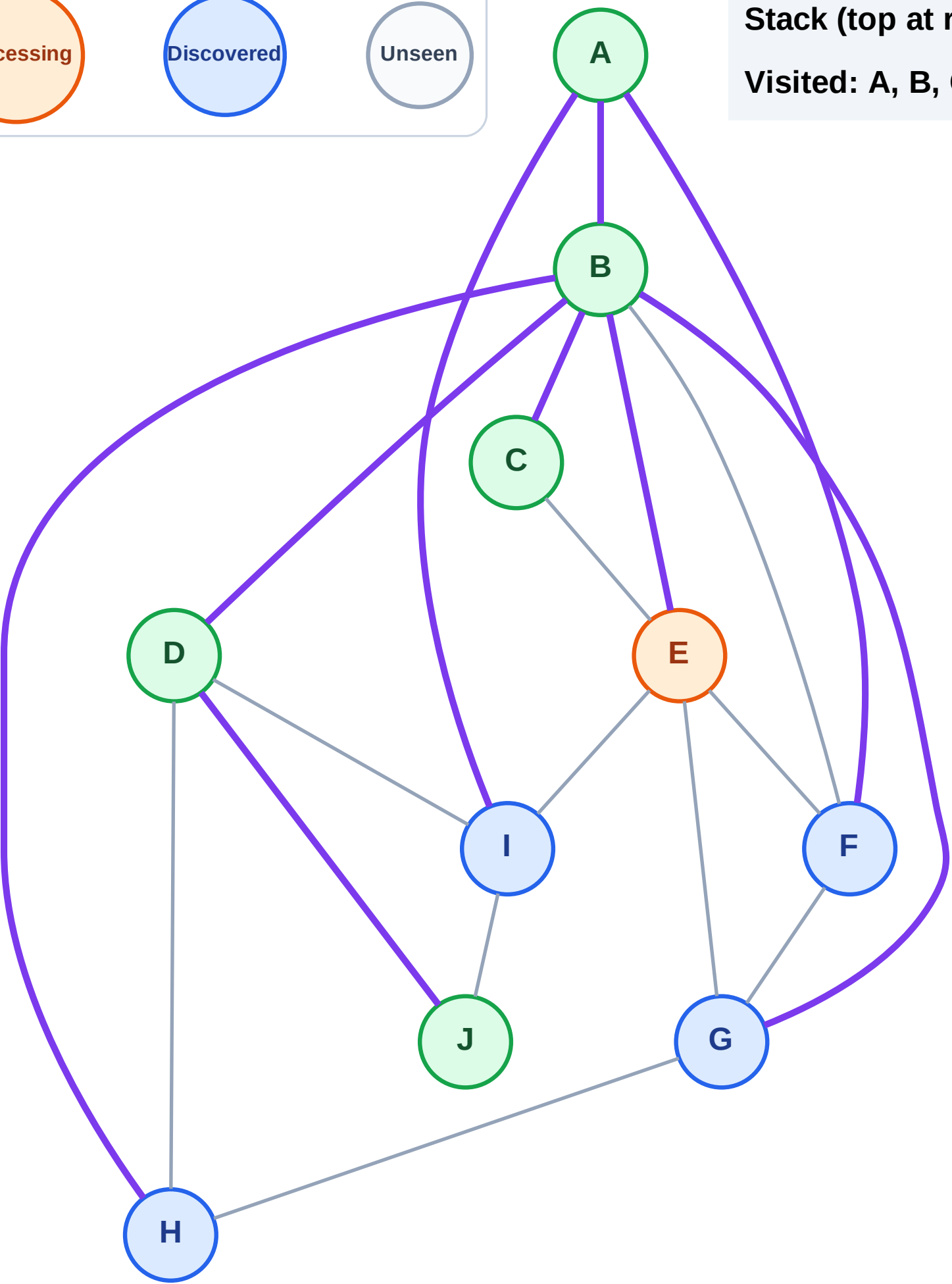
Step 12: Process node E

Legend



Stack (top at right): I | F | H | G

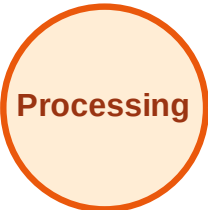
Visited: A, B, C, D, J



DFS Traversal

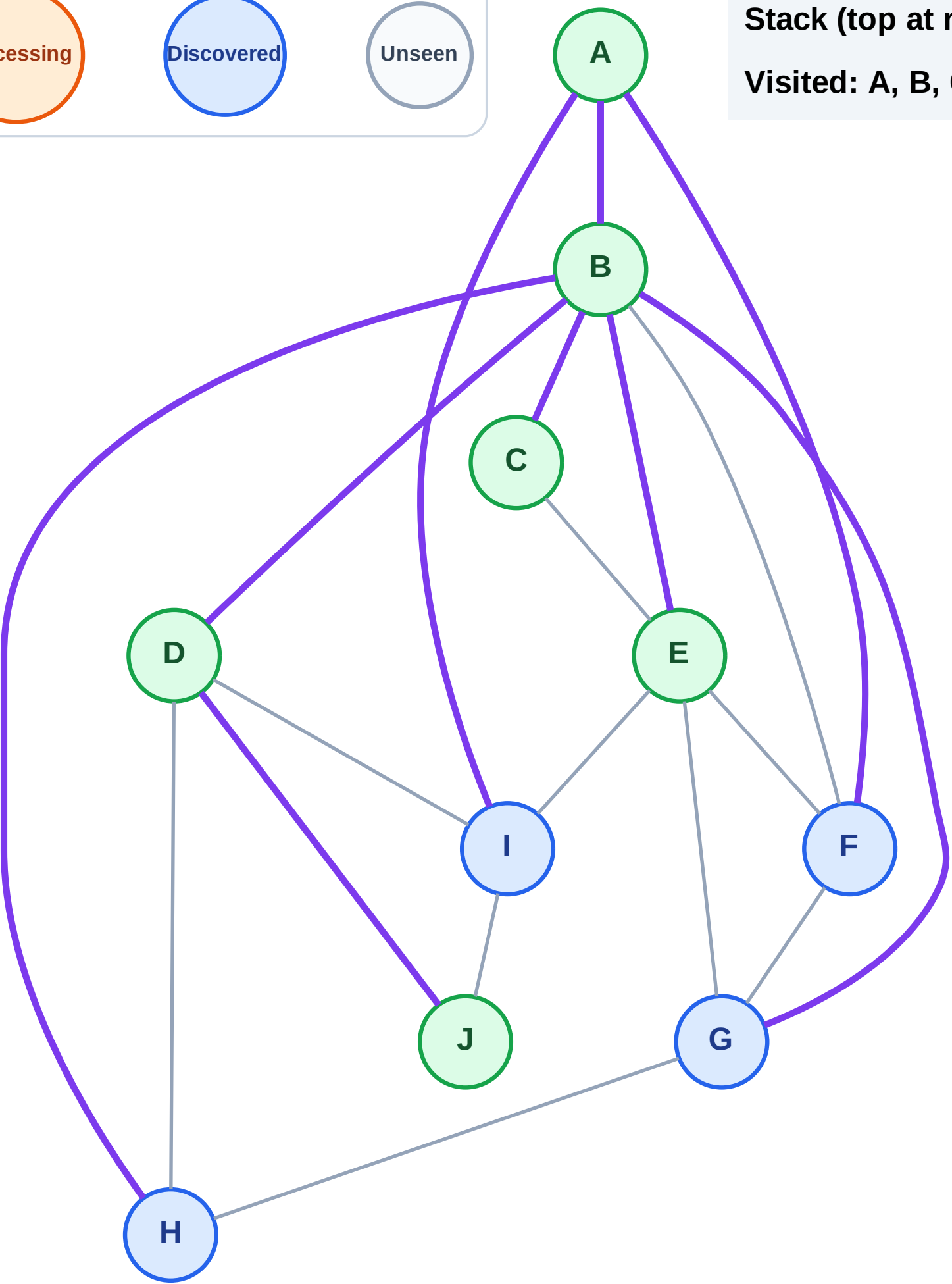
Step 13: No new neighbors from E

Legend



Stack (top at right): I | F | H | G

Visited: A, B, C, D, E, J



DFS Traversal

Step 14: Process node G

Legend

Complete

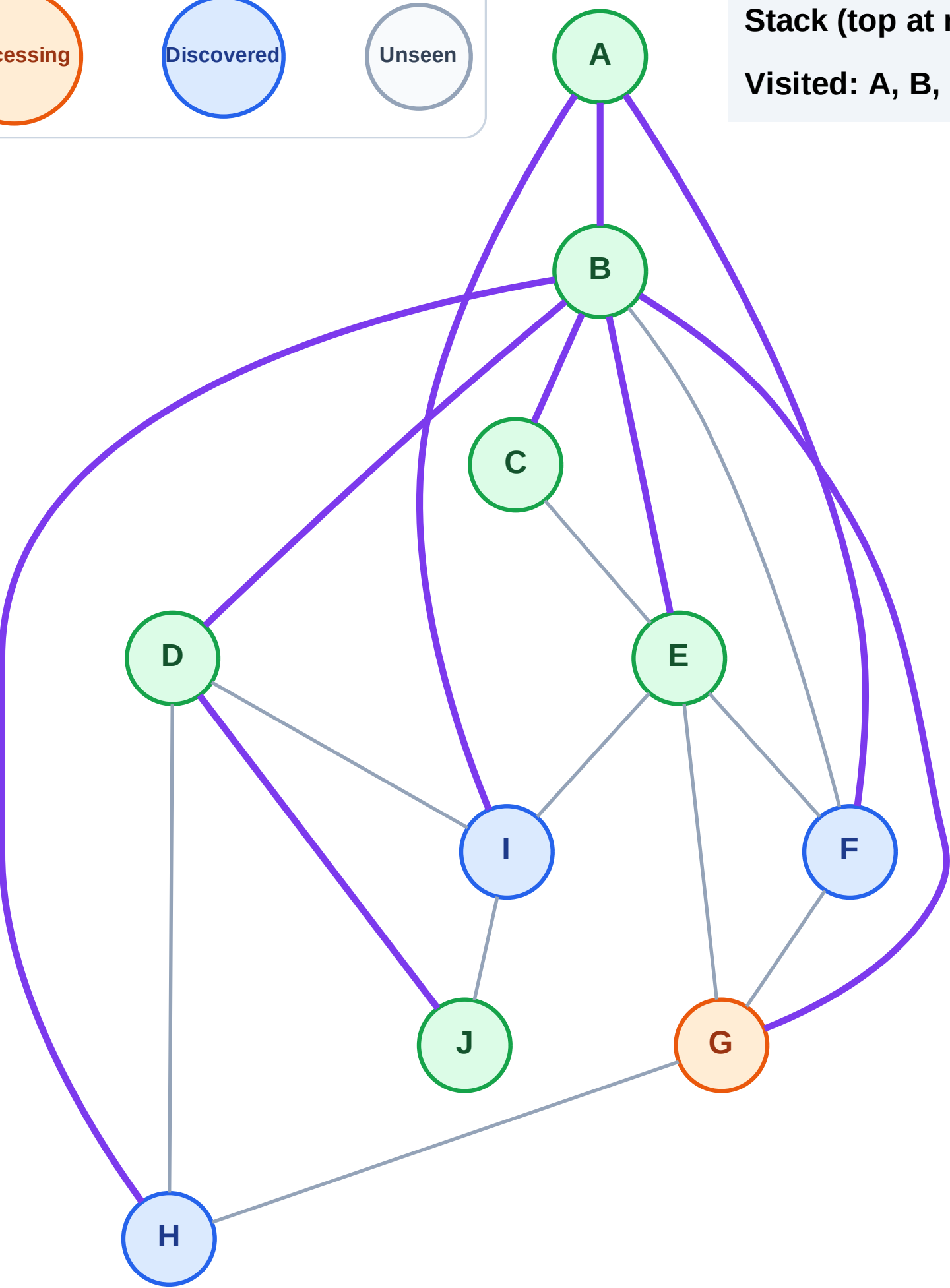
Processing

Discovered

Unseen

Stack (top at right): I | F | H

Visited: A, B, C, D, E, J

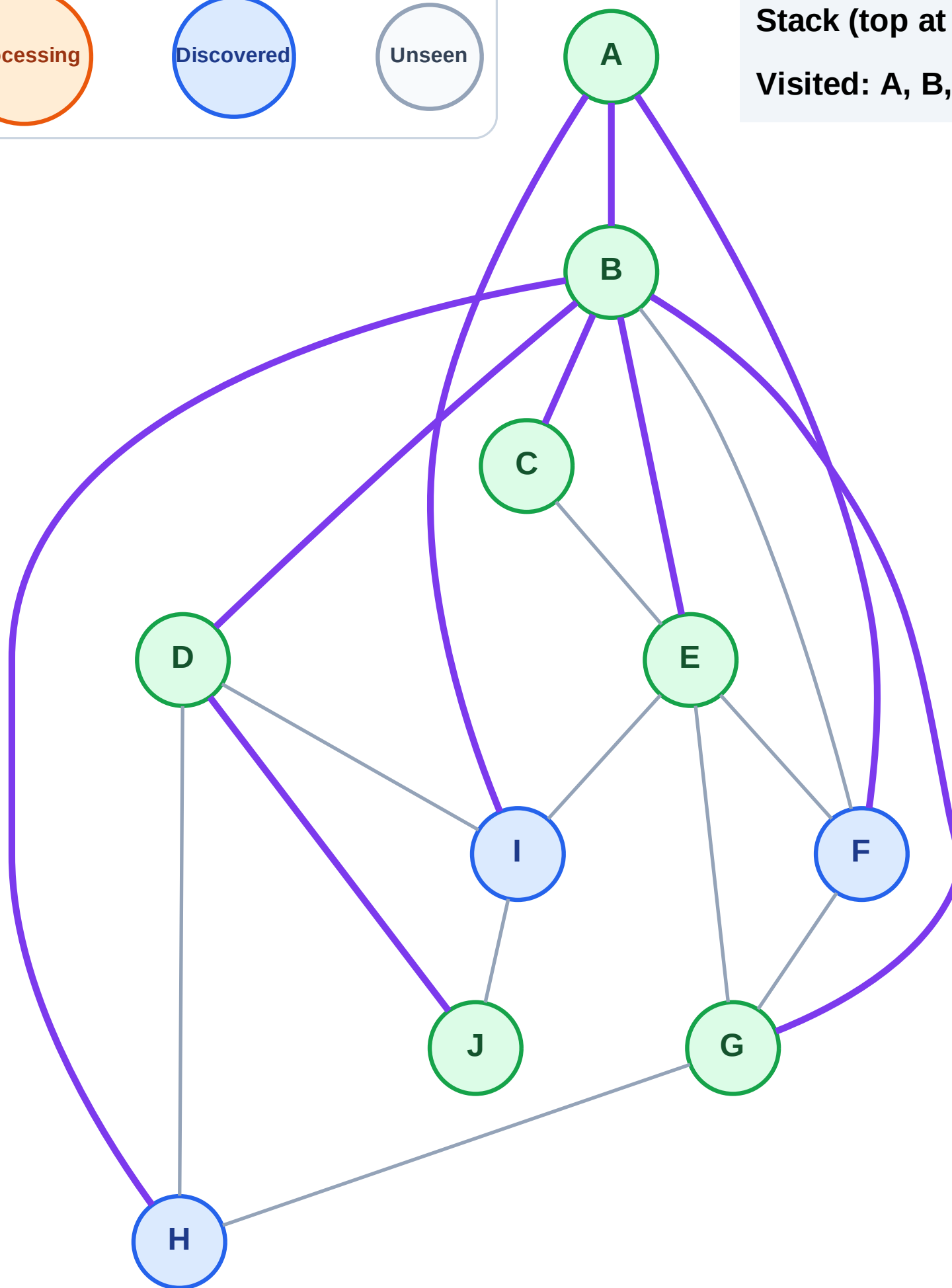


Step 15: No new neighbors from G

Legend

Unseen

Visited: A, B, C, D, E, G, J



DFS Traversal

Step 16: Process node H

Legend

Complete

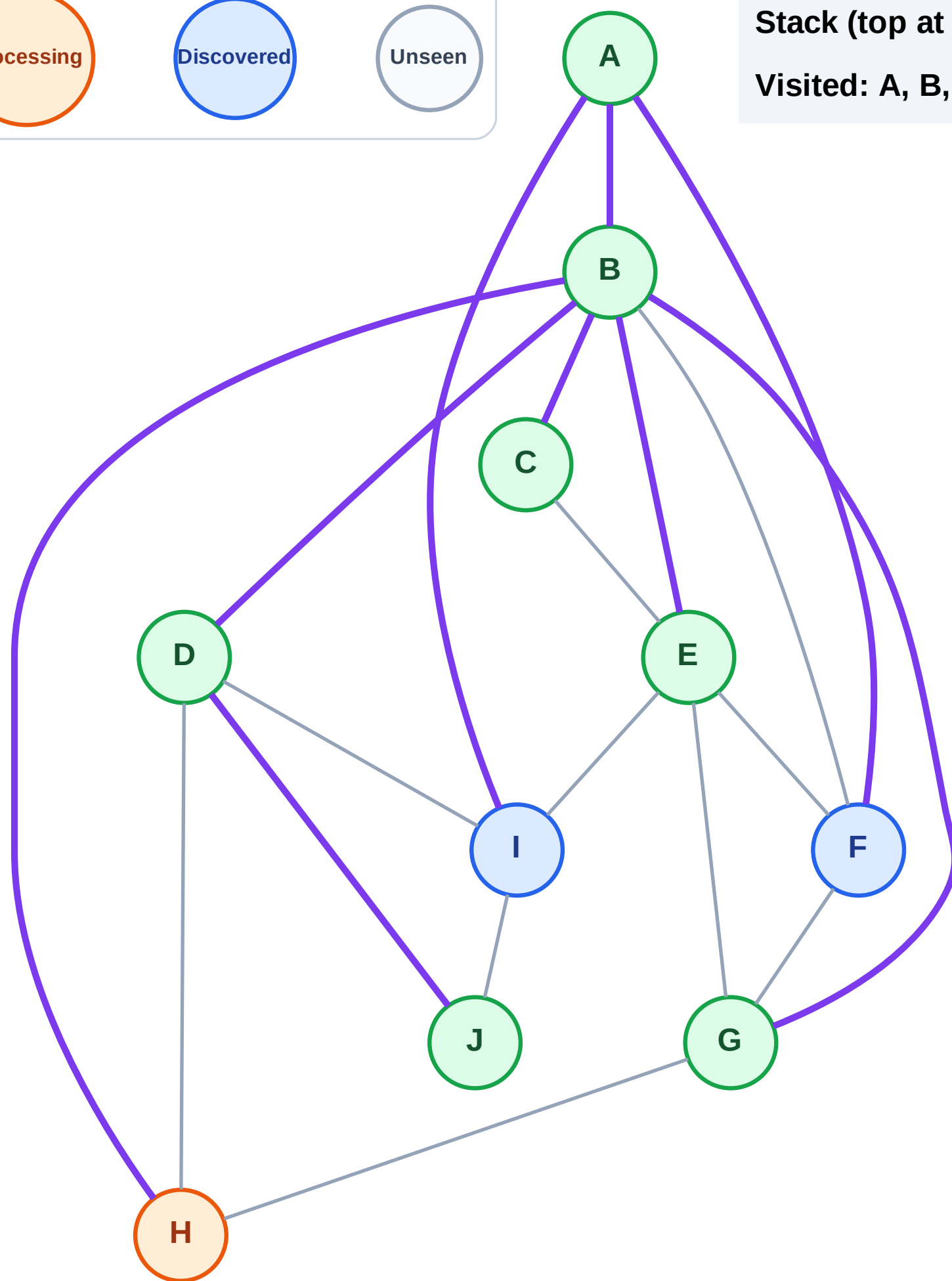
Processing

Discovered

Unseen

Stack (top at right): I | F

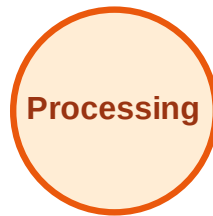
Visited: A, B, C, D, E, G, J



DFS Traversal

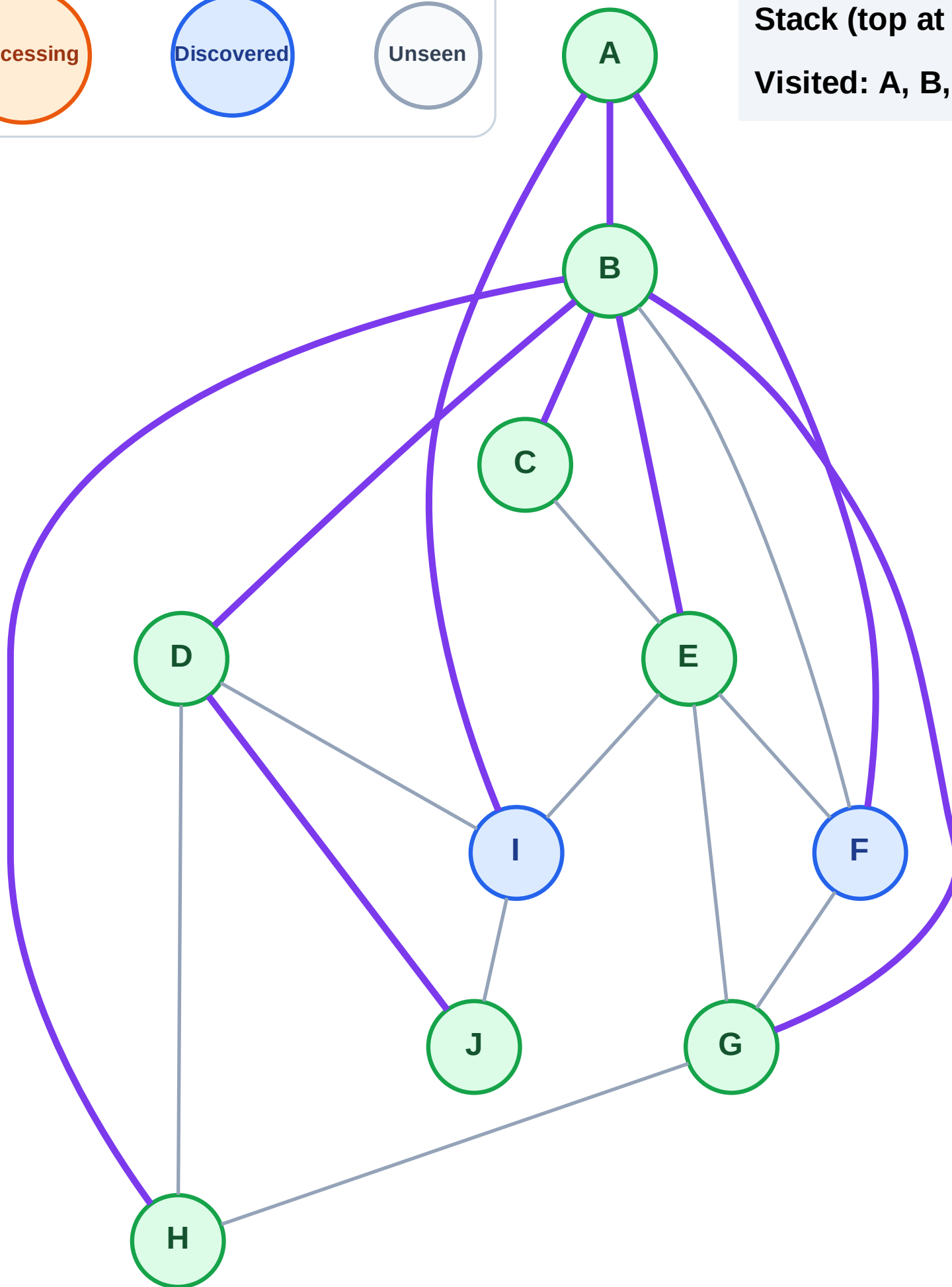
Step 17: No new neighbors from H

Legend



Stack (top at right): I | F

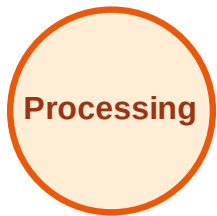
Visited: A, B, C, D, E, G, H, J



DFS Traversal

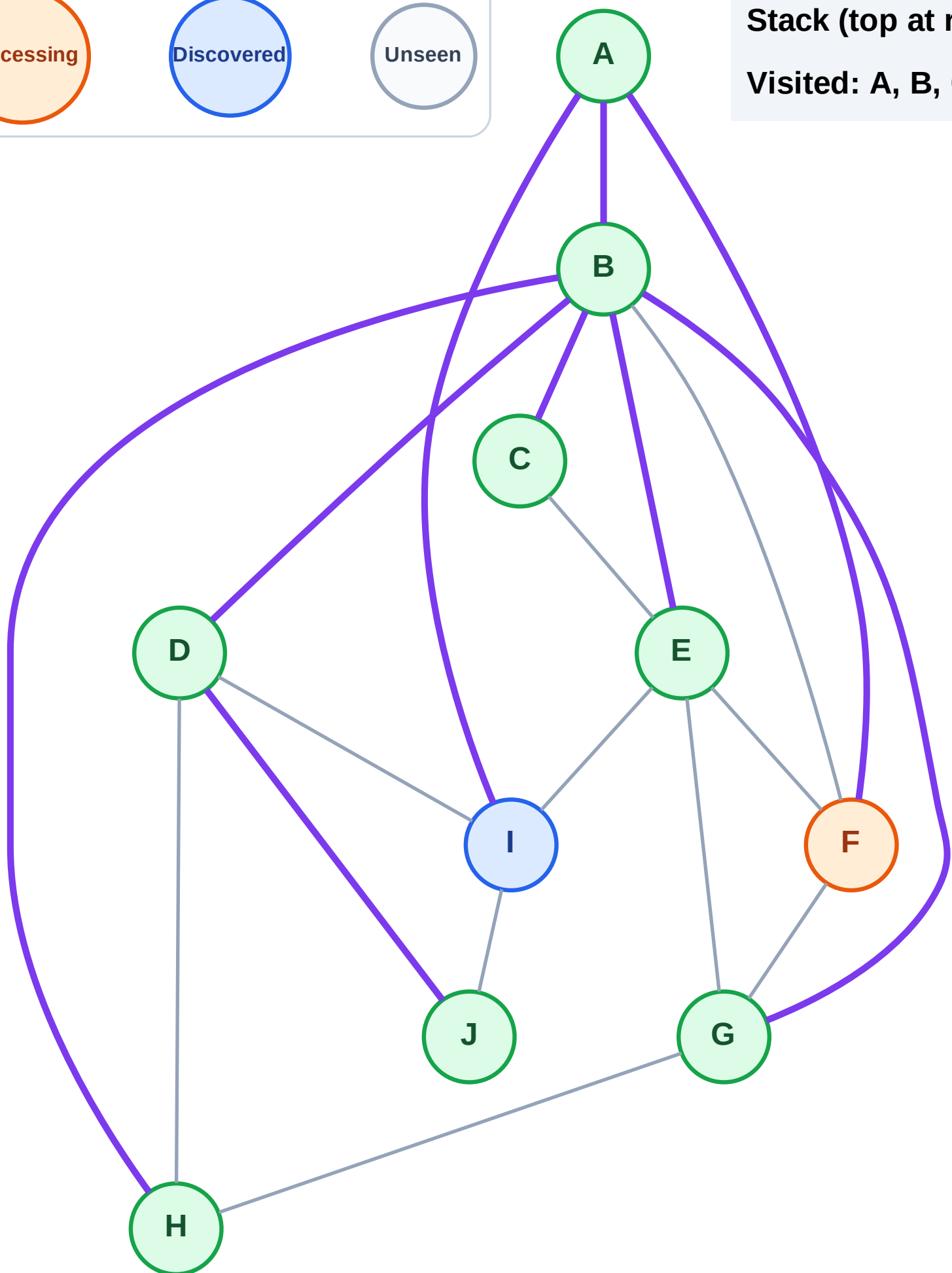
Step 18: Process node F

Legend



Stack (top at right): I

Visited: A, B, C, D, E, G, H, J

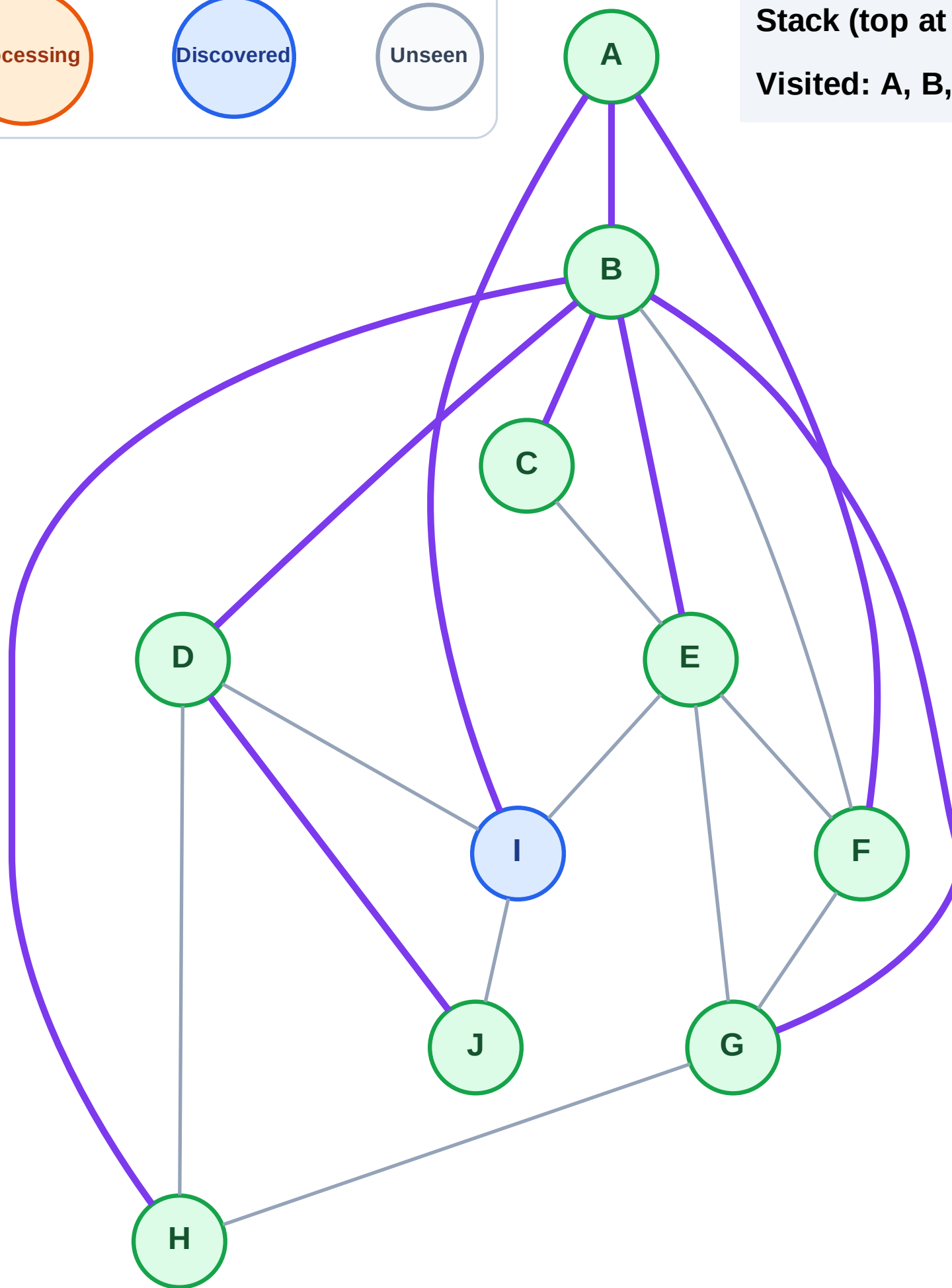


Step 19: No new neighbors from F

Legend

Unseen

Visited: A, B, C, D, E, F, G, H, J



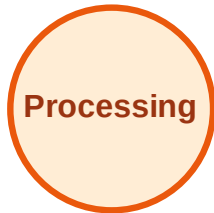
DFS Traversal

Step 20: Process node I

Legend



Complete



Processing



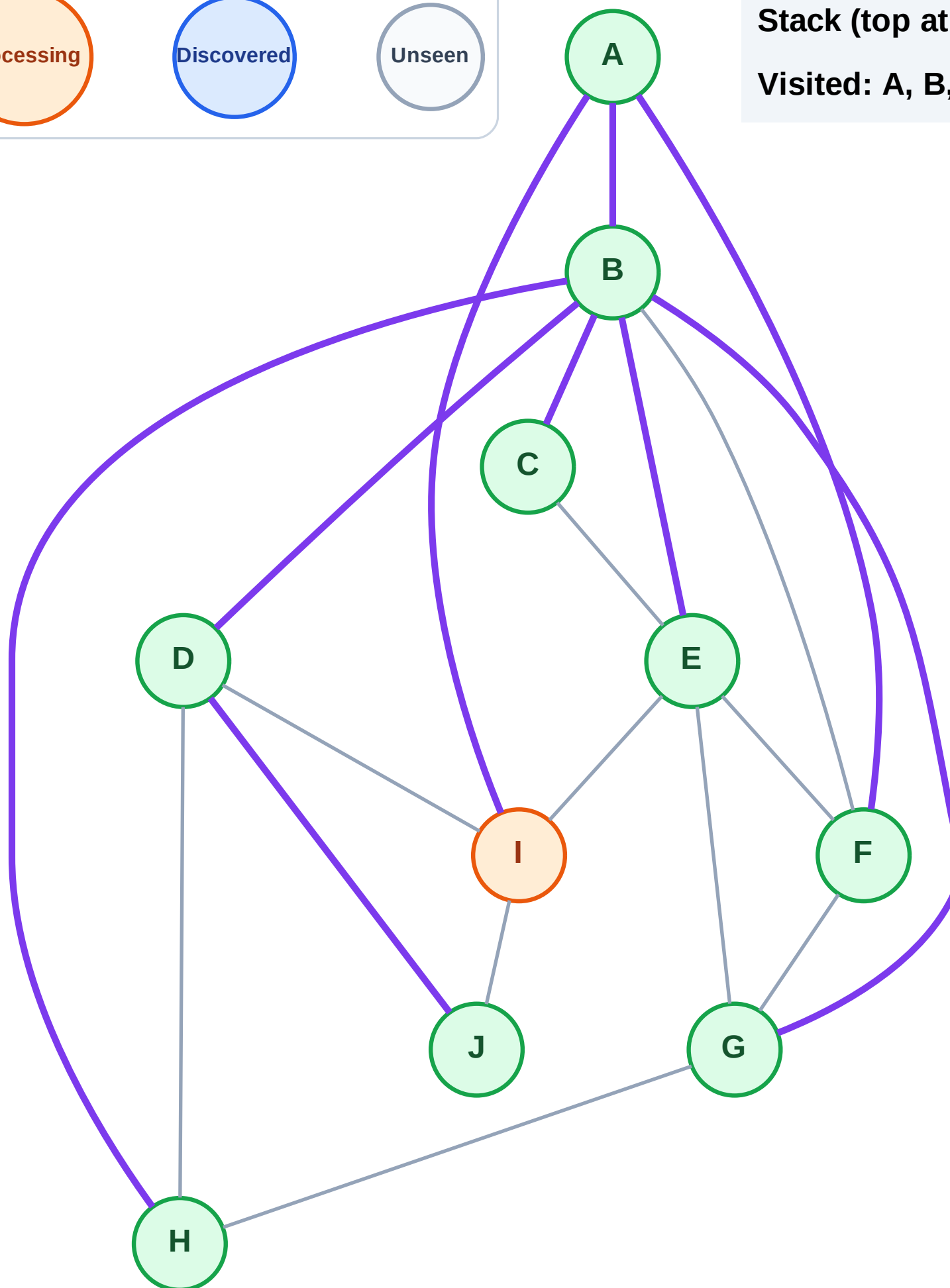
Discovered



Unseen

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

